

Clinical Embryology

Carle Illinois
**COLLEGE of
MEDICINE**



Wessam Ibrahim, MD, PhD, MEHP

(For use only by CIMED students)

Learning Objectives

- Differentiate between embryonic/fetal age & gestational age.
- Describe the major events of the first three weeks of human development, including fertilization, cleavage, implantation, and gastrulation.
- Explain the formation and derivatives of the three germ layers while correlating them with clinically relevant congenital anomalies.
- Describe the development and functions of extraembryonic membranes and the placenta during early pregnancy.
- Explain the establishment of maternal-fetal circulation and the endocrine functions of the placenta.
- Apply embryologic principles to interpret abnormalities of implantation and placentation, including ectopic pregnancy, placenta previa, and placenta accreta.

Learning Objectives

- Describe the embryologic origins of the gonads, genital ducts, and external genitalia.
- Explain the molecular and hormonal mechanisms of sex determination and sexual differentiation, including the roles of SRY and AMH.
- Differentiate the derivatives of the mesonephric (Wolffian) and paramesonephric (Müllerian) ducts in males and females.
- Analyze the embryologic basis of common congenital malformations of the reproductive system, including cryptorchidism, hypospadias, Müllerian anomalies, and disorders of sexual development.
- Integrate embryologic, placental, and reproductive developmental concepts to solve clinical and USMLE Step 1-style questions involving congenital abnormalities and reproductive disorders.

1

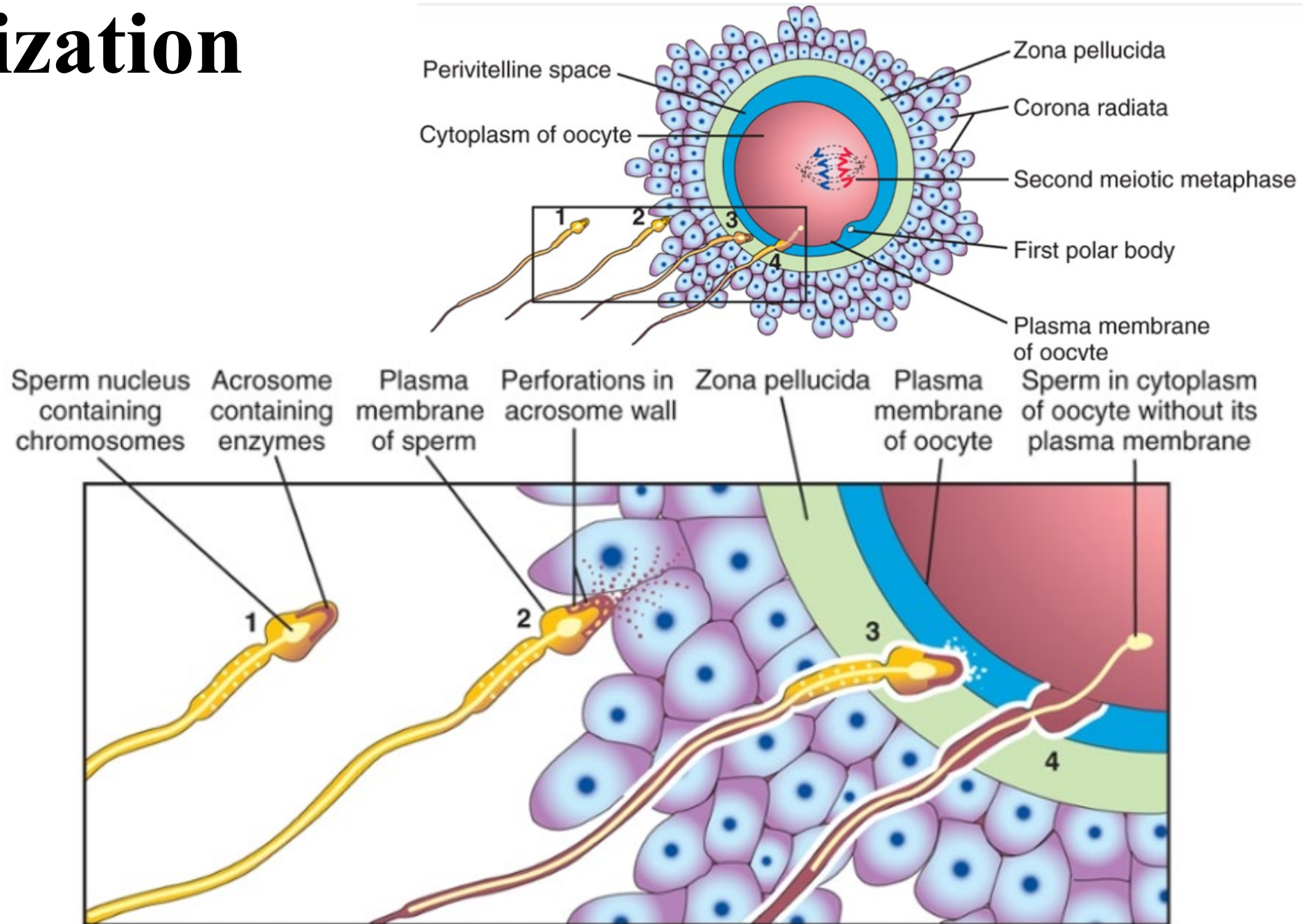


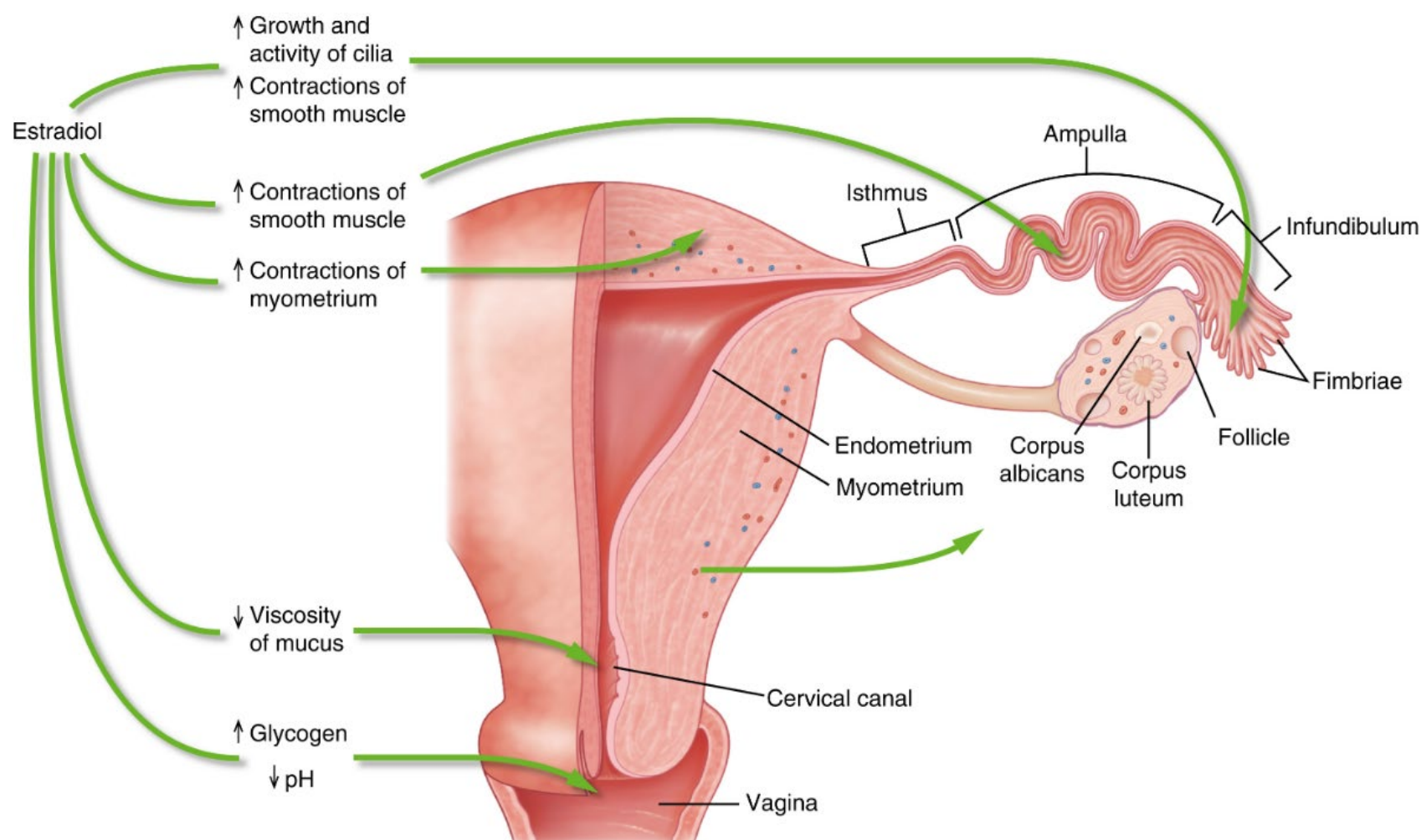
Question 1

A 30-year-old woman presents with 2 years of unexplained infertility. Evaluation of her partner reveals normal sperm concentration, morphology, and motility. During assisted reproductive testing, sperm successfully reach and surround the secondary oocyte but are unable to bind effectively to the zona pellucida or undergo the acrosome reaction. Further studies demonstrate failure of the biochemical changes that normally occur as sperm transit through the female reproductive tract. Which of the following processes is most likely defective in this patient?

- A. Completion of meiosis II by the secondary oocyte
- B. Capacitation of spermatozoa
- C. Cortical granule exocytosis
- D. Formation of the male pronucleus
- E. Fusion of the male and female pronuclei

Fertilization





Capacitation is the step in the maturation of sperm that is required to fertilize an oocyte. This step is a biochemical event; the sperm move normally and look mature prior to capacitation. *In vivo*, capacitation occurs after ejaculation, when the sperms leave the vagina and enter the female reproductive system. The uterus aids in the steps of capacitation by lipoproteins, proteolytic enzymes, and glycosidase enzymes.

Source : Figure in Drake, Vogl, Mitchell, Tibbitts, & Richardson (2015). *Gray's Atlas of anatomy* (2nd ed., p. 242). Elsevier.

Question 1

A 32-year-old woman presents with 2 years of unexplained infertility. Evaluation of her partner reveals normal sperm concentration, morphology, and motility. During assisted reproductive testing, sperm successfully reach and surround the secondary oocyte but are unable to bind effectively to the zona pellucida or undergo the acrosome reaction. Further studies demonstrate failure of the biochemical changes that normally occur as sperm transit through the female reproductive tract. Which of the following processes is most likely defective in this patient?

- A. Completion of meiosis II by the secondary oocyte
- B. Capacitation of spermatozoa
- C. Cortical granule exocytosis
- D. Formation of the male pronucleus
- E. Fusion of the male and female pronuclei

B

Explanation

Capacitation is a functional maturation process that occurs within the female reproductive tract.

During this process, sperm undergo membrane and metabolic changes that enable binding to the zona pellucida and initiation of the acrosome reaction.

Without capacitation, sperm may exhibit normal count, morphology, motility, and transport through the reproductive tract, but they cannot penetrate the zona pellucida and fertilize the oocyte.

Takeaway:

Capacitation → Acrosome reaction → Zona penetration → Cortical reaction → Completion of meiosis II → Pronuclear fusion.

Failure of capacitation is a classic cause of infertility despite otherwise normal semen analysis.

2



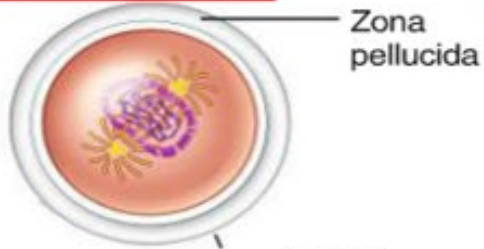
Question 2

A 34-year-old woman undergoes evaluation for recurrent implantation failure following multiple cycles of in vitro fertilization. Genetic testing demonstrates a normal embryo karyotype, and embryo development progresses appropriately to the blastocyst stage. Endometrial biopsy reveals inadequate invasion of maternal endometrial tissue by trophoblastic cells, preventing establishment of normal maternal-fetal exchange. Which of the following embryonic cell populations is primarily responsible for invasion of the endometrium and establishment of implantation?

- A. Cytotrophoblast
- B. Epiblast
- C. Syncytiotrophoblast
- D. Hypoblast
- E. Extraembryonic mesoderm

Cleavage

(a) Zygote
(fertilized egg)



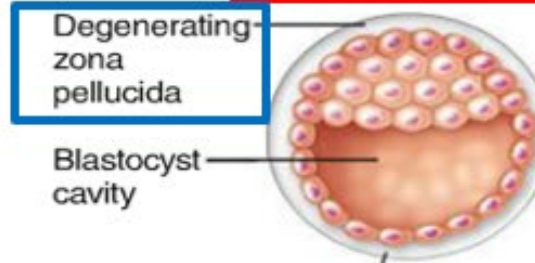
(b) 4-cell stage
2 days



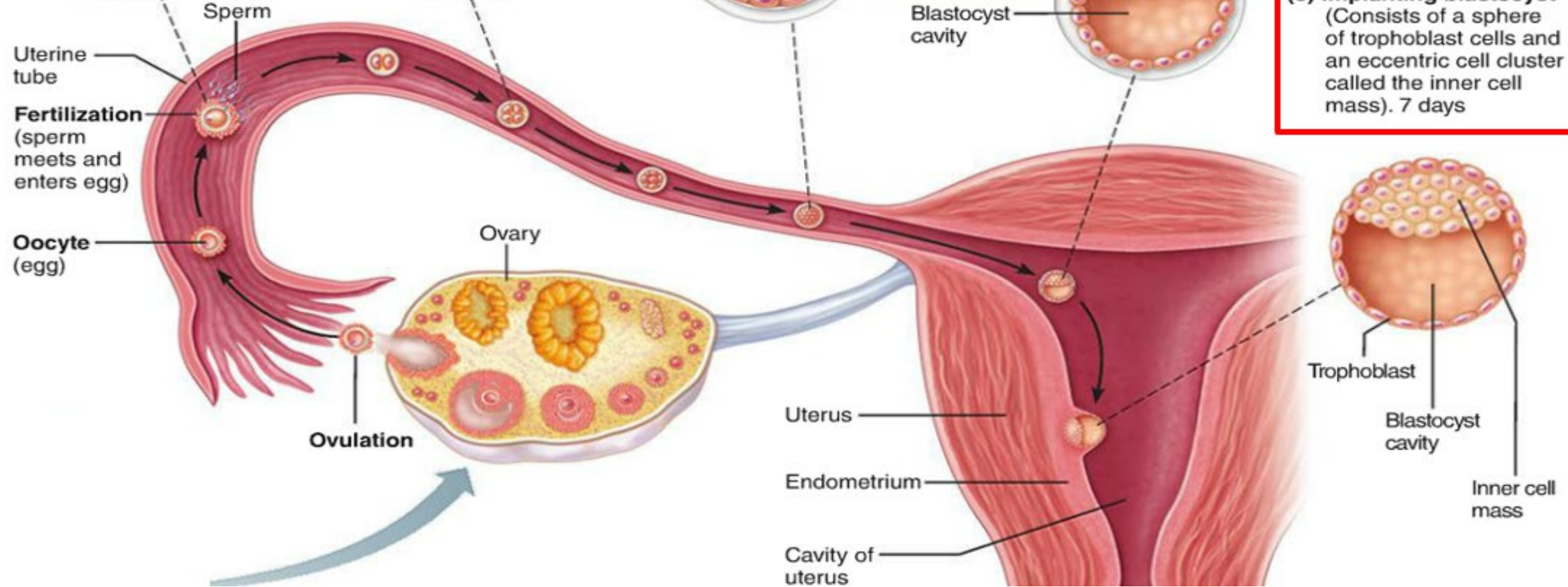
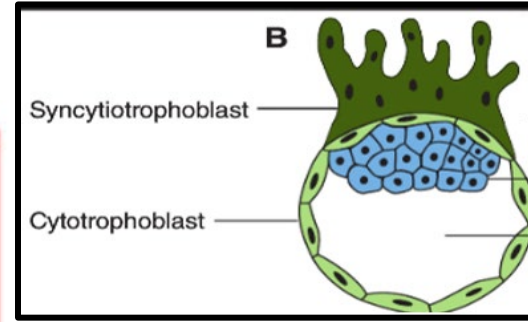
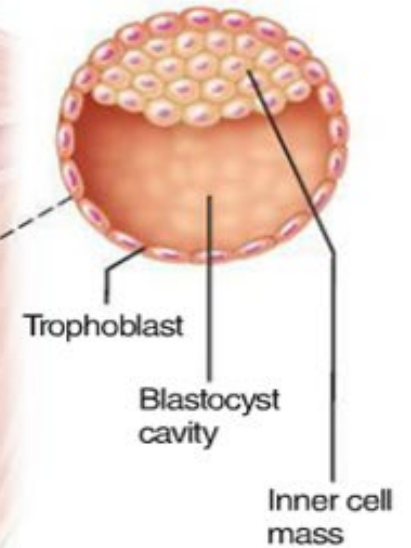
(c) Morula (a solid ball of blastomeres).
3 days



(d) Early blastocyst
(Morula hollows out, fills with fluid, and "hatches" from the zona pellucida).
4 days



(e) Implanting blastocyst
(Consists of a sphere of trophoblast cells and an eccentric cell cluster called the inner cell mass). 7 days



Question 2

A 34-year-old woman undergoes evaluation for recurrent implantation failure following multiple cycles of in vitro fertilization. Genetic testing demonstrates a normal embryo karyotype, and embryo development progresses appropriately to the blastocyst stage. Endometrial biopsy reveals inadequate invasion of maternal endometrial tissue by trophoblastic cells, preventing establishment of normal maternal-fetal exchange. Which of the following embryonic cell populations is primarily responsible for invasion of the endometrium and establishment of implantation?

- A. Cytotrophoblast
- B. Epiblast
- C. Syncytiotrophoblast
- D. Hypoblast
- E. Extraembryonic mesoderm

C

Explanation

Following implantation, the trophoblast differentiates into two layers:

- **Cytotrophoblast:** the inner cellular layer that retains mitotic activity and gives rise to new trophoblastic cells.
- **Syncytiotrophoblast:** the outer multinucleated layer that invades the endometrium, erodes maternal capillaries, and establishes the early uteroplacental circulation.
- The syncytiotrophoblast also secretes **human chorionic gonadotropin (hCG)**, which **maintains the corpus luteum during early pregnancy**. Failure of syncytiotrophoblast invasion can lead to implantation failure, early pregnancy loss, and abnormal placentation.

Takeaway:

- **Syncytiotrophoblast** = implantation + invasion + hCG production.
- **A useful association is:**

Cytotrophoblast → proliferates

Syncytiotrophoblast → invades and secretes hCG

3



Question 3

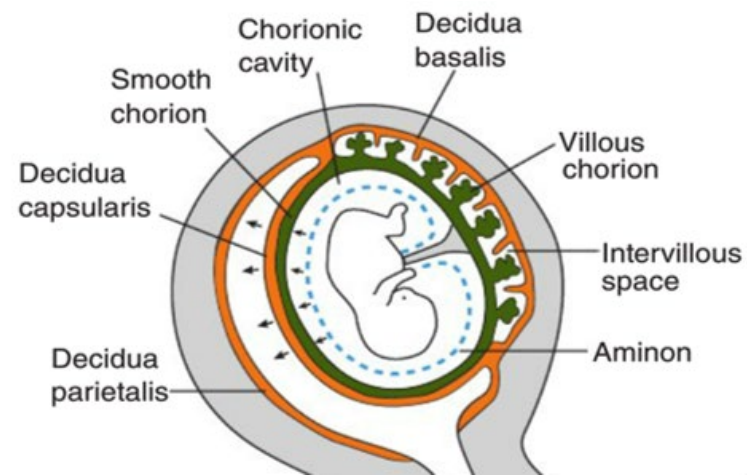
A 38-year-old woman, gravida 4 para 3, presents in labor at 39 weeks' gestation. Her obstetric history is significant for two prior cesarean deliveries. Following an uncomplicated vaginal delivery, the placenta fails to separate spontaneously. Attempts at manual extraction result in massive postpartum hemorrhage requiring emergent surgical intervention. Histologic examination demonstrates chorionic villi abnormally attached directly to the myometrium. Which of the following is the most likely diagnosis?

- A. Placenta accreta
- B. Placental abruption
- C. Placenta previa
- D. Velamentous placenta
- E. Membranous placenta

Placenta

Parts:

- Maternal → Decidua basalis.
- Fetal → chorionic villi (chorion frondosum)



NORMAL PREGNANCY

The placenta attaches to a temporary layer in the uterus that's shed at delivery



PLACENTA ACCRETA

When the placenta attaches too deeply into the uterine wall



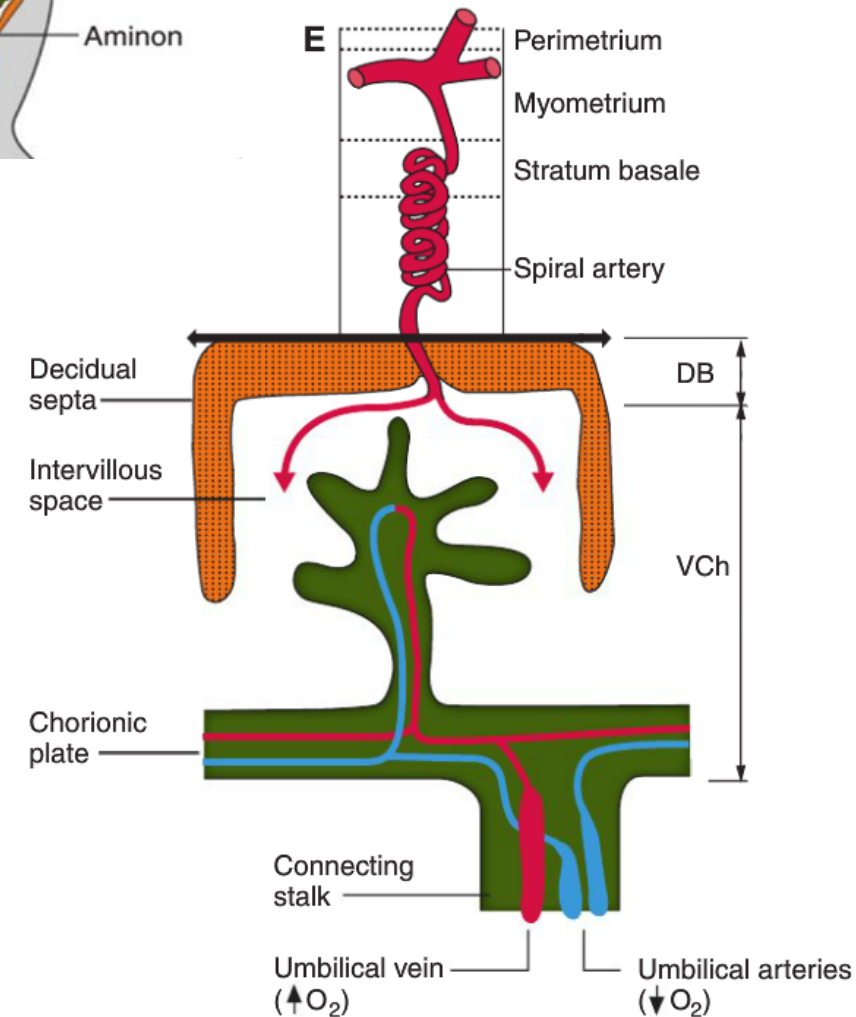
PLACENTA INCRETA

When the placenta attaches into the uterine muscle



PLACENTA PERCRETA

When the placenta goes completely through the uterine wall, sometimes invading nearby organs like the bladder



Placental abnormalities

- **Placenta previa** occurs when the placenta attaches in the lower part of the uterus, covering the internal os.
- **Placental abruption** occurs when a normally implanted placenta prematurely separates from the uterus before delivery of the fetus. It is associated with maternal hypertension.
- **Placental accreta** occurs when there is abnormal adherence of the chorionic villi to the uterine wall with partial or complete absence of the decidua basalis.
- **Placental percreta** occurs when the chorionic villi penetrate the myometrium to reach the perimetrium

Placenta Accreta

Placenta grows too deep into the wall of the uterus



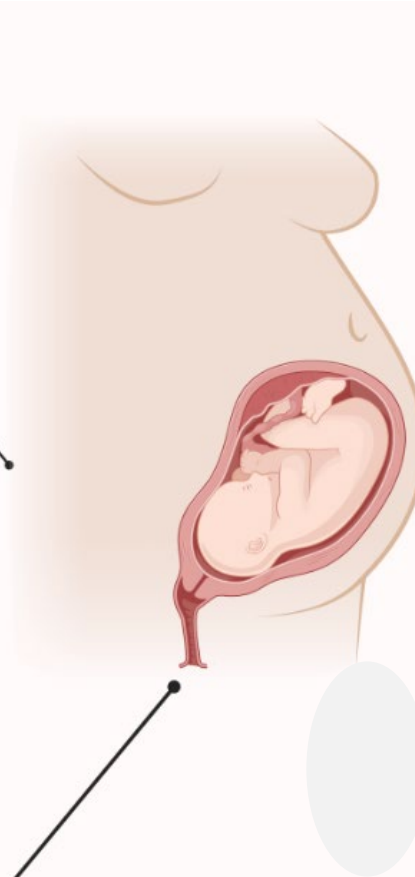
Placenta Previa

Placenta attaches very low in the uterus



Placenta Abruption

Placenta prematurely separates from the uterus



Question 3

A 38-year-old woman, gravida 4 para 3, presents in labor at 39 weeks' gestation. Her obstetric history is significant for two prior cesarean deliveries. Following an uncomplicated vaginal delivery, the placenta fails to separate spontaneously. Attempts at manual extraction result in massive postpartum hemorrhage requiring emergent surgical intervention. Histologic examination demonstrates chorionic villi abnormally attached directly to the myometrium.

Which of the following is the most likely diagnosis?

- A. Placenta accreta
- B. Placental abruption
- C. Placenta previa
- D. Velamentous placenta
- E. Membranous placenta

A

Explanation

Placenta accreta occurs when there is defective formation of the decidua basalis, allowing chorionic villi to attach directly to the myometrium. This abnormal adherence prevents normal placental separation after delivery and can result in life-threatening postpartum hemorrhage. Risk factors include: **previous cesarean delivery, placenta previa, prior uterine surgery, multiparity.**

- **Placenta previa:** placenta overlies or is near the internal cervical os and presents with painless third-trimester vaginal bleeding before delivery.
- **Placental abruption:** premature separation of a normally implanted placenta and presents with painful vaginal bleeding, uterine tenderness, and fetal distress.
- **Velamentous placenta:** umbilical vessels travel within the membranes before reaching the placenta and is associated with vasa previa and fetal hemorrhage.
- **Membranous placenta** (placenta membranacea): rare condition in which villi persist over a large portion of the chorion and does not characteristically cause failure of placental separation after delivery.

4

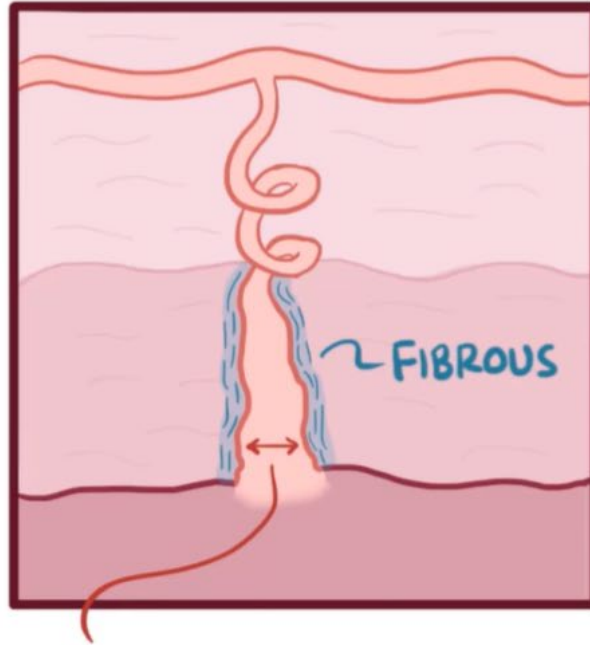


Question 4

A 28-year-old primigravid woman at 29 weeks' gestation presents with a 2-week history of worsening headaches, lower-extremity edema, and visual disturbances. Her blood pressure is 170/110 mm Hg, and urinalysis demonstrates 4+ proteinuria. Fetal ultrasonography reveals growth restriction. Following delivery, placental histopathology demonstrates inadequate remodeling of the maternal spiral arteries. Failure of which of the following developmental processes most likely contributed to this patient's condition?

- A. Formation of the bilaminar embryonic disc
- B. Migration of neural crest cells into the pharyngeal arches
- C. Trophoblastic invasion of maternal vessels
- D. Gastrulation and formation of the trilaminar embryonic disc
- E. Development of the allantois and urachus

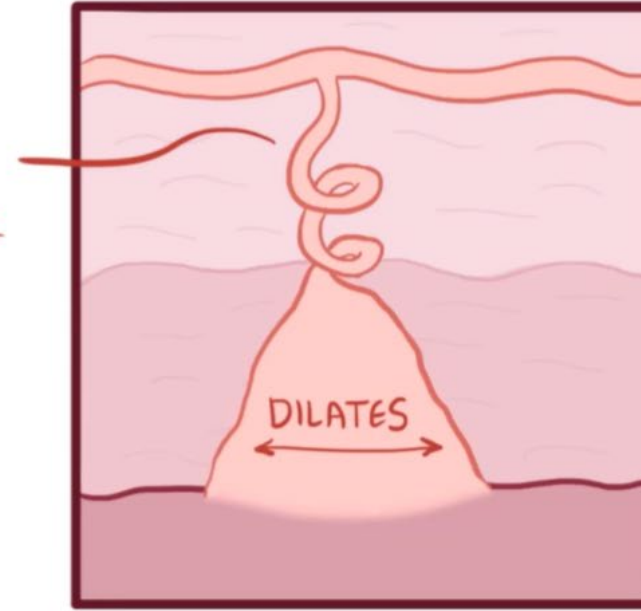
PREECLAMPSIA



NARROW ~ LESS BLOOD

NORMAL

SPIRAL ARTERIES
* DELIVER
LOTS of
BLOOD



Osmosis.org

OSMOSIS.org
2023 Edition

Normal placentation: Trophoblast invasion → Spiral artery remodeling → Low-resistance uteroplacental circulation

Preeclampsia: Defective trophoblast invasion → High-resistance spiral arteries → Placental ischemia → Hypertension and proteinuria. Therefore, defective **trophoblastic invasion of maternal vessels** is the embryologic process most directly responsible for this condition.

Question 4

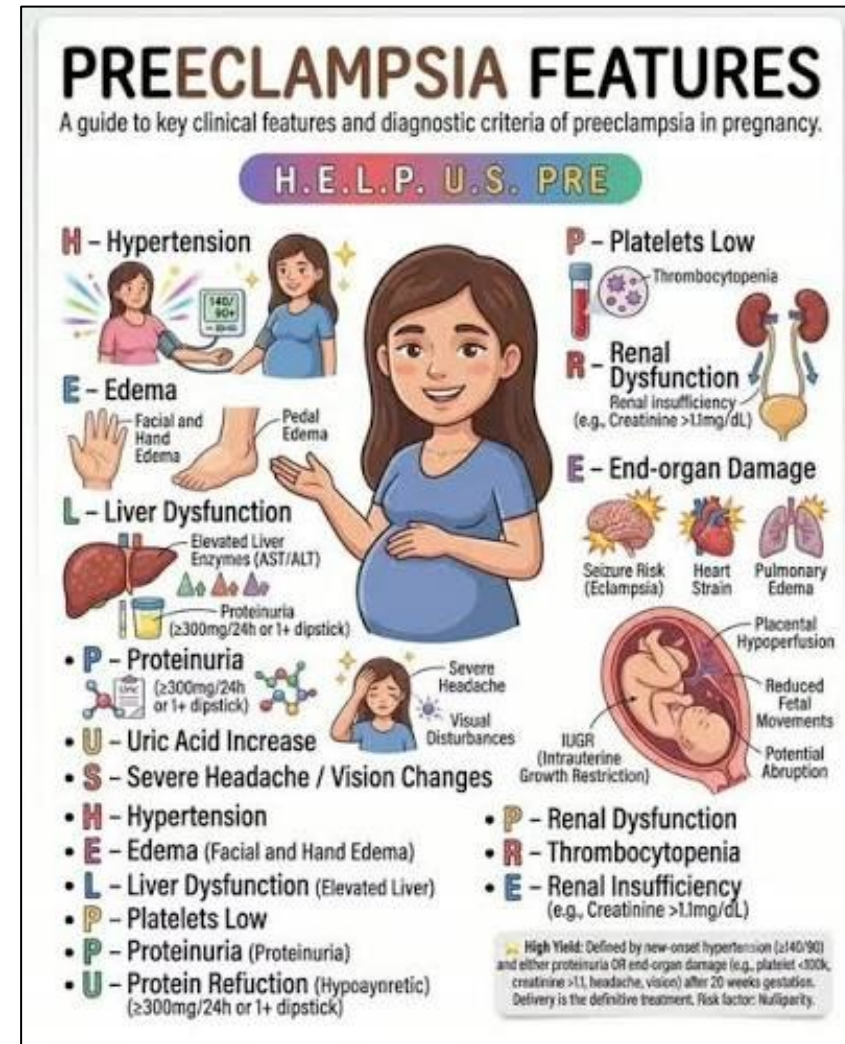
A 28-year-old primigravid woman at 29 weeks' gestation presents with a 2-week history of worsening headaches, lower-extremity edema, and visual disturbances. Her blood pressure is 170/110 mm Hg, and urinalysis demonstrates 4+ proteinuria. Fetal ultrasonography reveals growth restriction. Following delivery, placental histopathology demonstrates inadequate remodeling of the maternal spiral arteries. Failure of which of the following developmental processes most likely contributed to this patient's condition?

- A. Formation of the bilaminar embryonic disc
- B. Migration of neural crest cells into the pharyngeal arches
- C. Trophoblastic invasion of maternal vessels
- D. Gastrulation and formation of the trilaminar embryonic disc
- E. Development of the allantois and urachus

C

Explanation

- Preeclampsia is caused by **abnormal placentation**, specifically inadequate invasion of the maternal spiral arteries by **trophoblasts**.
- During normal development, trophoblastic cells invade and remodel the spiral arteries, converting them from small, high-resistance vessels into large, low-resistance vessels. Failure of this trophoblastic invasion results in poor placental perfusion and the release of factors that cause widespread maternal endothelial dysfunction, leading to **hypertension, proteinuria, and end-organ injury** characteristic of preeclampsia.



5

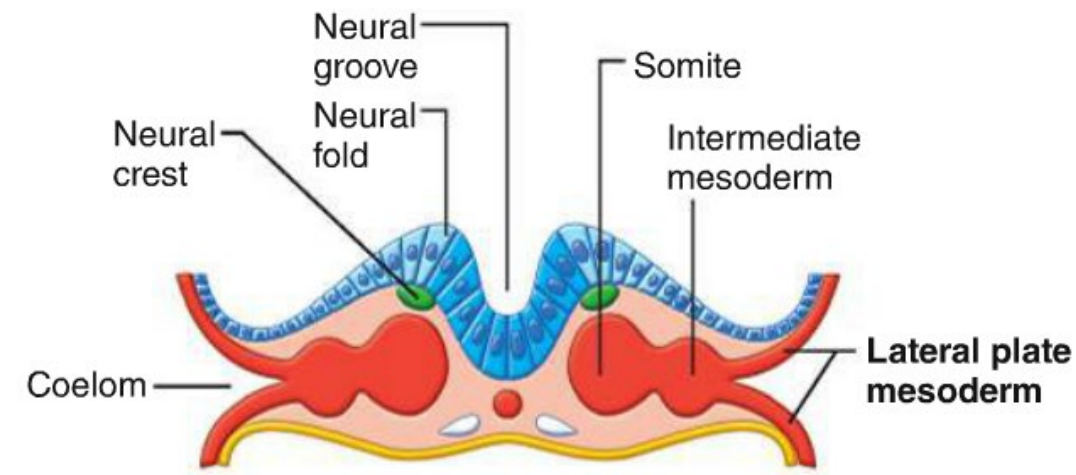
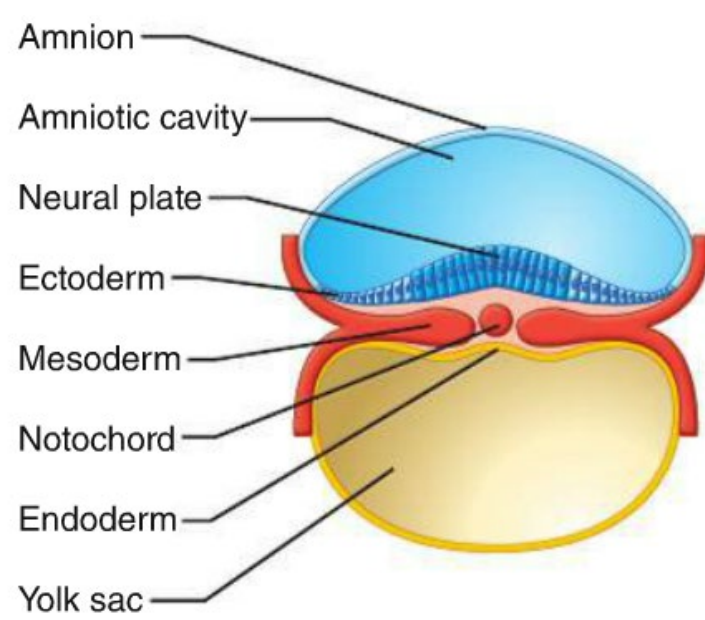
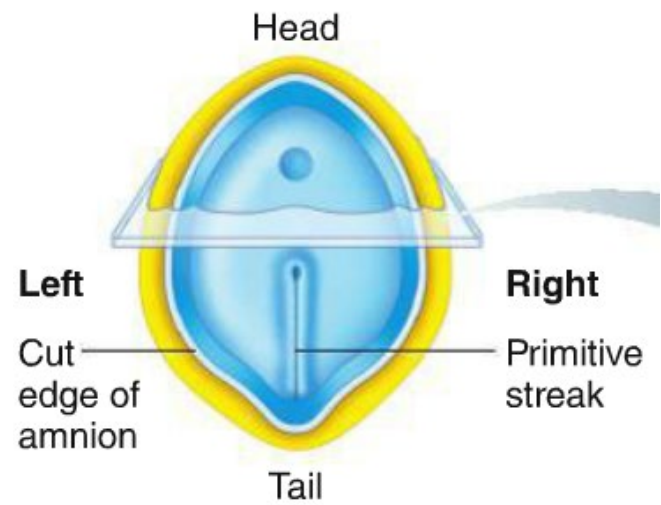


Question 5

A newborn phenotypic female is evaluated for primary gonadal failure. Genetic testing reveals a 46-XX karyotype. Pelvic ultrasonography demonstrates small, poorly developed ovaries with a marked deficiency of germ cells. The physician explains that normal ovarian development depends on the migration of primordial germ cells during early embryogenesis. Failure of this migration can result in gonadal dysgenesis and infertility. Primordial germ cells normally migrate to the developing gonadal ridges from which of the following structures?

- A. Cloaca
- B. Mesonephric duct
- C. Paramesonephric duct
- D. Yolk Sac
- E. Genital ridge

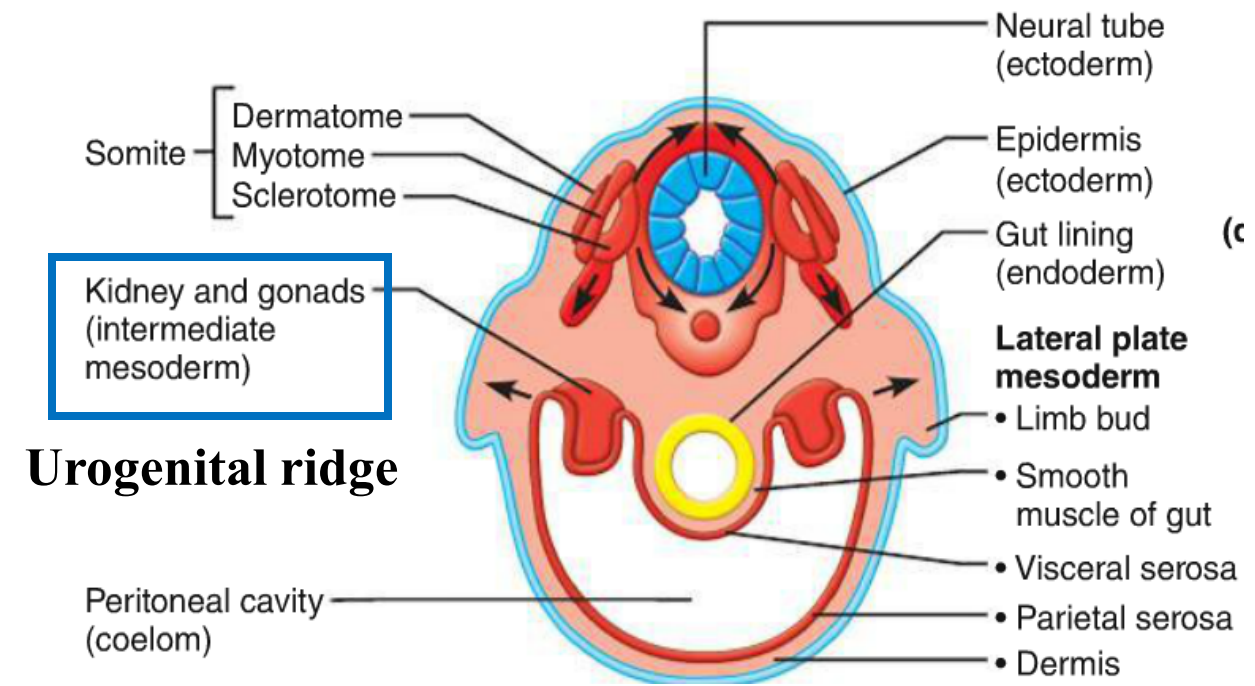
Mesodermal Differentiation



Urogenital ridge

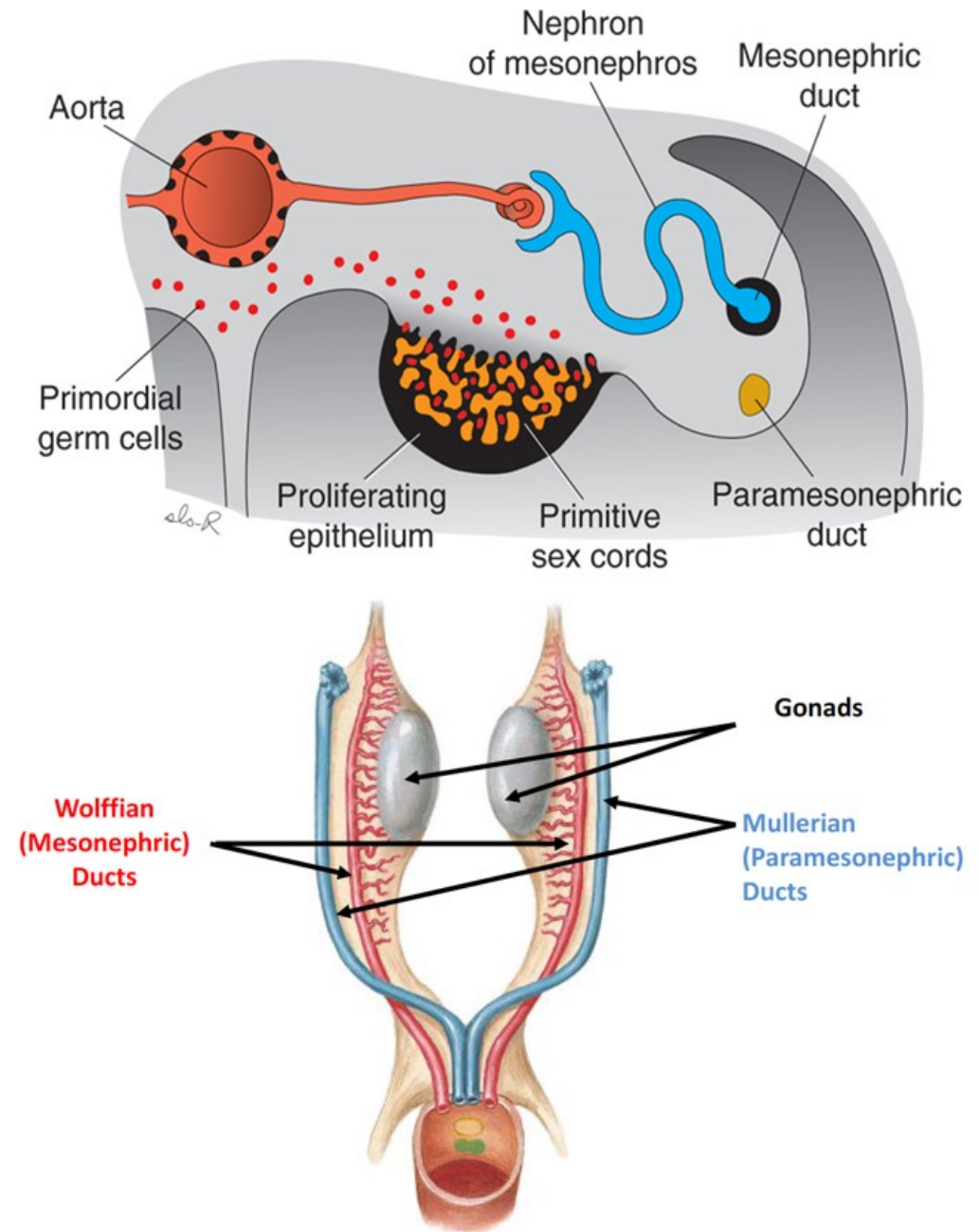
Nephrogenic cord

Genital ridge



The Reproductive Systems

- The components include gonads, ducts & external genitalia from mesoderm and endoderm.
 - The genotype is determined at fertilization (XX, XY).
 - **Week 7** embryo begins phenotypic sexual differentiation.
 - The presence of the Y chromosome determines sexual dimorphism. **The germ cells have an inductive influence on the differentiation of the gonad.**
 - **Week 12** external genitalia recognizable
 - **Week 20** phenotypic differentiation complete.
- At gastrulation, the primordial germ cells migrate to the yolk sac. During week 6, they migrate to the genital ridge.**



Question 5

A newborn phenotypic female is evaluated for primary gonadal failure. Genetic testing reveals a 46,XX karyotype. Pelvic ultrasonography demonstrates small, poorly developed ovaries with a marked deficiency of germ cells. The physician explains that normal ovarian development depends on the migration of primordial germ cells during early embryogenesis. Failure of this migration can result in gonadal dysgenesis and infertility. Primordial germ cells normally migrate to the developing gonadal ridges from which of the following structures?

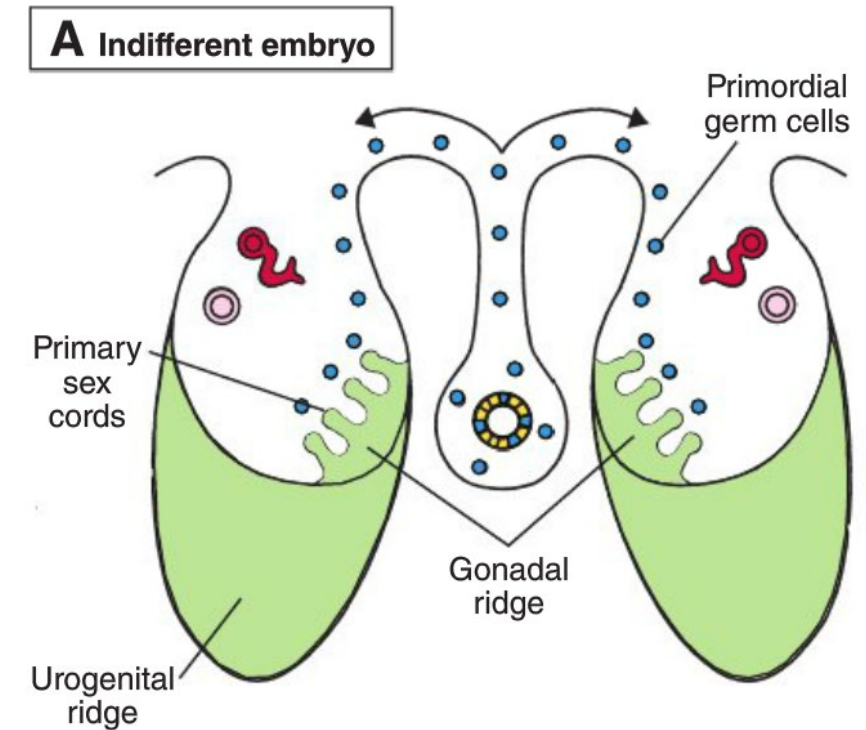
- A. Cloaca
- B. Mesonephric duct
- C. Paramesonephric duct
- D. Yolk Sac
- E. Genital ridge

D

Explanation

Primordial germ cells (PGCs) are the precursors of spermatozoa and oocytes.

They originate in the **wall of the yolk sac near the allantois** and subsequently migrate through the dorsal mesentery to the developing **gonadal ridges** during weeks 4 to 6 of development. Once they reach the gonads, they differentiate into oogonia or spermatogonia. Failure of this migration results in a marked reduction or absence of germ cells within the gonads, leading to **gonadal dysgenesis and infertility**.



6



Question 6

A newborn is evaluated for ambiguous genitalia. Karyotype analysis demonstrates 46,XY chromosomes. Pelvic ultrasonography reveals the absence of a uterus and fallopian tubes. Genetic testing identifies a mutation in the SRY gene that prevents expression of its protein product during embryogenesis. Failure of expression of the SRY gene product would most directly disrupt which of the following developmental processes?

- A. Descent of the testes into the scrotum
- B. Regression of the mesonephric (Wolffian) ducts
- C. Conversion of testosterone to dihydrotestosterone (DHT)
- D. Fusion of the paramesonephric (Müllerian) ducts
- E. Differentiation of Sertoli cells within the primitive sex cords

Sex Differentiation

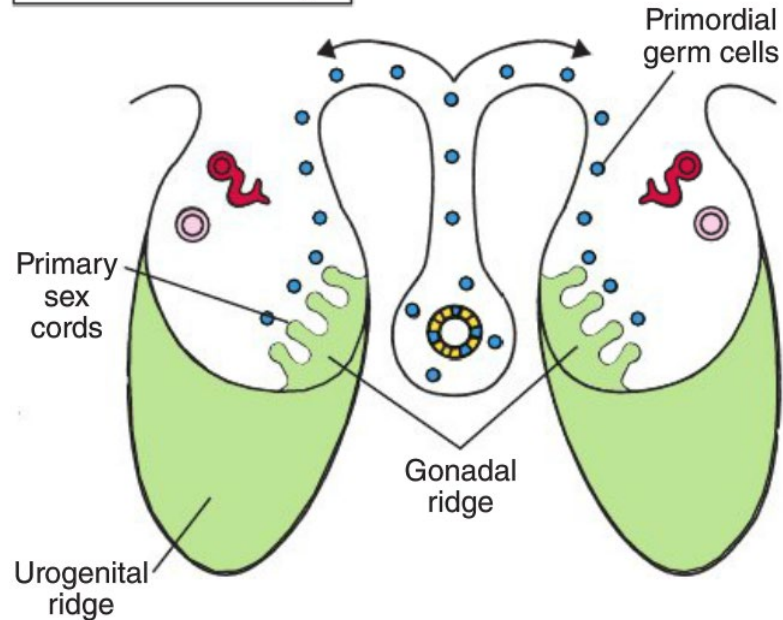
SIGNAL	SOURCE	ACTION	RESULT IF ABSENT
SRY gene	Y chromosome	Induces indifferent gonad → testis (Sertoli + Leydig cells)	Gonad becomes ovary (default) NRE
AMH (Anti-Müllerian Hormone)	Sertoli cells of testis	Causes regression of Müllerian (paramesonephric) ducts	Müllerian ducts persist → uterus + tubes + upper vagina form NRE
Testosterone	Leydig cells of testis	Stabilises Wolffian (mesonephric) ducts → male reproductive tract	Wolffian ducts regress → no vas deferens, epididymis, seminal vesicles NRE
DHT (Dihydrotestosterone)	Testosterone converted by 5 α -reductase in target tissues	Masculinises external genitalia (penis, scrotum, prostate)	Female external genitalia form (default) — even in XY individuals NRE

- **Female development is the default pathway:** it occurs in the absence of SRY gene expression and without requiring estrogen.
- Male development requires active hormonal signaling: SRY → testes → testosterone + AMH.
- Remove the testes (or block testosterone/AMH) and female structures develop regardless of karyotype.

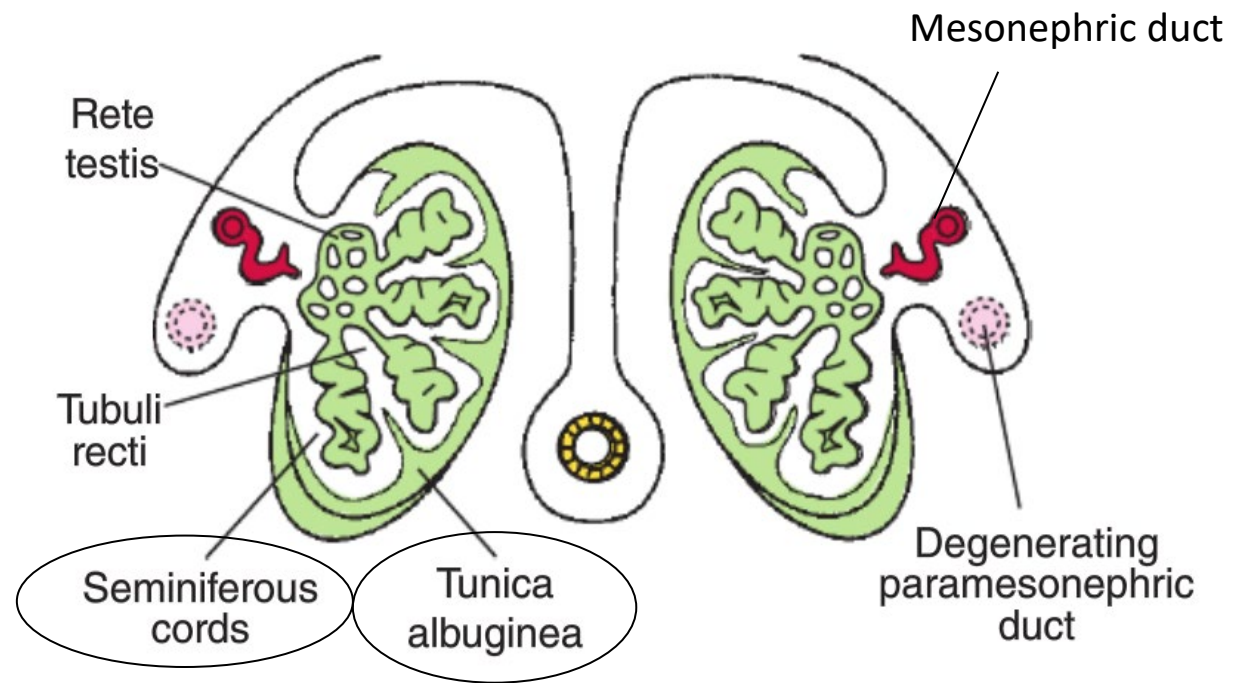
Gonad differentiation: Testis

SRY gene on Y chromosome encodes the testes-determining factor (TDF)

A Indifferent embryo



C Week 20



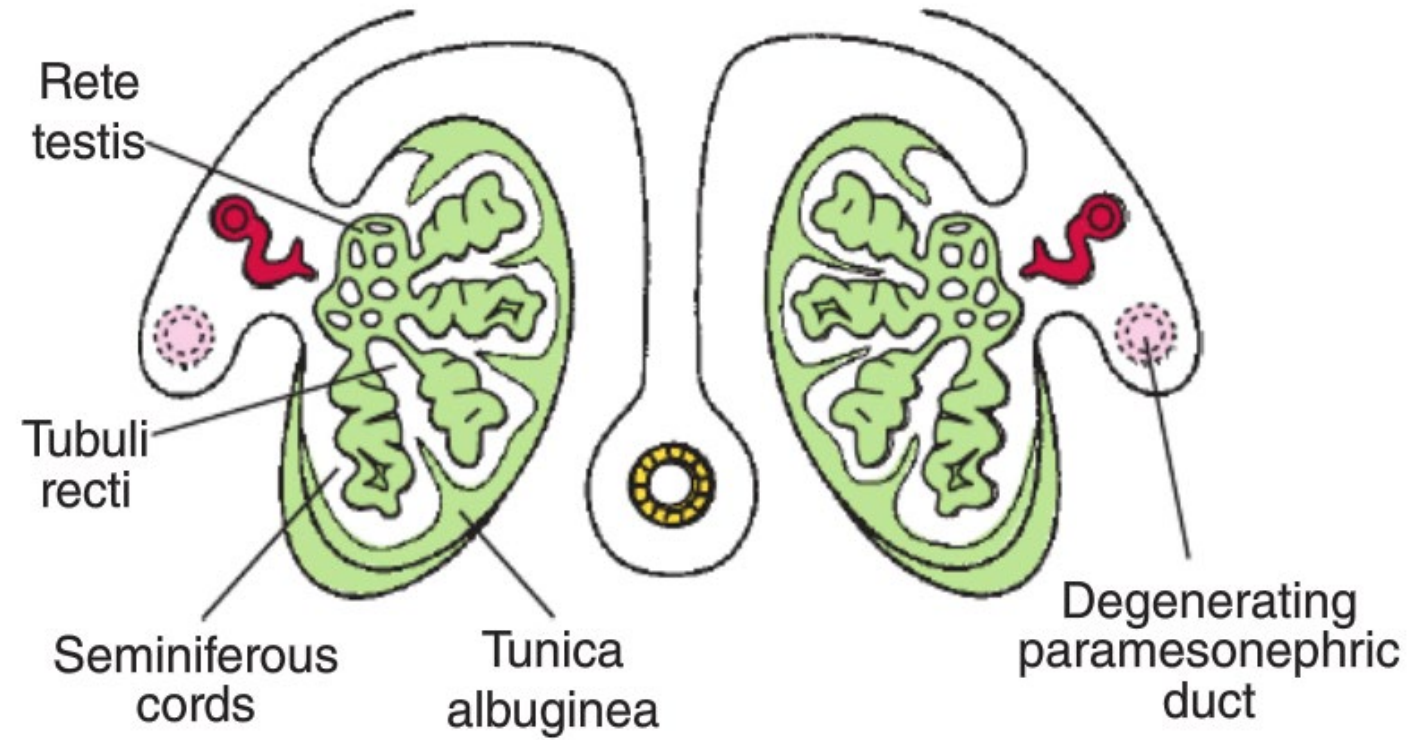
After the germ cells arrive, the gonads will differentiate into testes, beginning with the formation of the seminiferous tubules from the sex cords and the tunica albuginea as a thickened capsule around the testes.



Development of the Testis

Sertoli cells, in testes, produce **anti-Mullerian hormone (AMH)** or **Mullerian Inhibiting Substance (MIS)** causes the degeneration of the paramesonephric ducts (Mullerian Ducts).

Leydig cells, in the testes, produce **testosterone**. Testosterone affects mesonephric ducts and external genitalia development.



Question 6

A newborn is evaluated for ambiguous genitalia. Karyotype analysis demonstrates 46,XY chromosomes. Pelvic ultrasonography reveals the absence of a uterus and fallopian tubes. Genetic testing identifies a mutation in the SRY gene that prevents expression of its protein product during embryogenesis. Failure of expression of the SRY gene product would most directly disrupt which of the following developmental processes?

- A. Descent of the testes into the scrotum
- B. Regression of the mesonephric (Wolffian) ducts
- C. Conversion of testosterone to dihydrotestosterone (DHT)
- D. Fusion of the paramesonephric (Müllerian) ducts
- E. Differentiation of Sertoli cells within the primitive sex cords

E

Explanation

- The SRY (Sex-determining Region of the Y chromosome) gene is the testis-determining gene located on the short arm of the Y chromosome. Its protein product functions as a transcription factor that initiates differentiation of the indifferent gonad into a testis.
- The earliest major effect of SRY expression is the development of Sertoli cells from the primitive sex cords. Sertoli cells subsequently secrete anti-Müllerian hormone (AMH), causing regression of the Müllerian ducts.

A developmental sequence to remember is:

- SRY expression → Testis formation → Leydig cell differentiation → Testosterone production → Wolffian duct development.
- SRY expression → Sertoli cell differentiation → AMH secretion → Müllerian duct regression

7



Question 7

A 19-year-old male is evaluated for infertility. Physical examination is unremarkable. Semen analysis demonstrates markedly decreased sperm volume and absent fructose. Pelvic MRI reveals bilateral absence of the seminal vesicles, while the testes and epididymides are present and normally developed. The absent structures are derived from which embryologic structure?

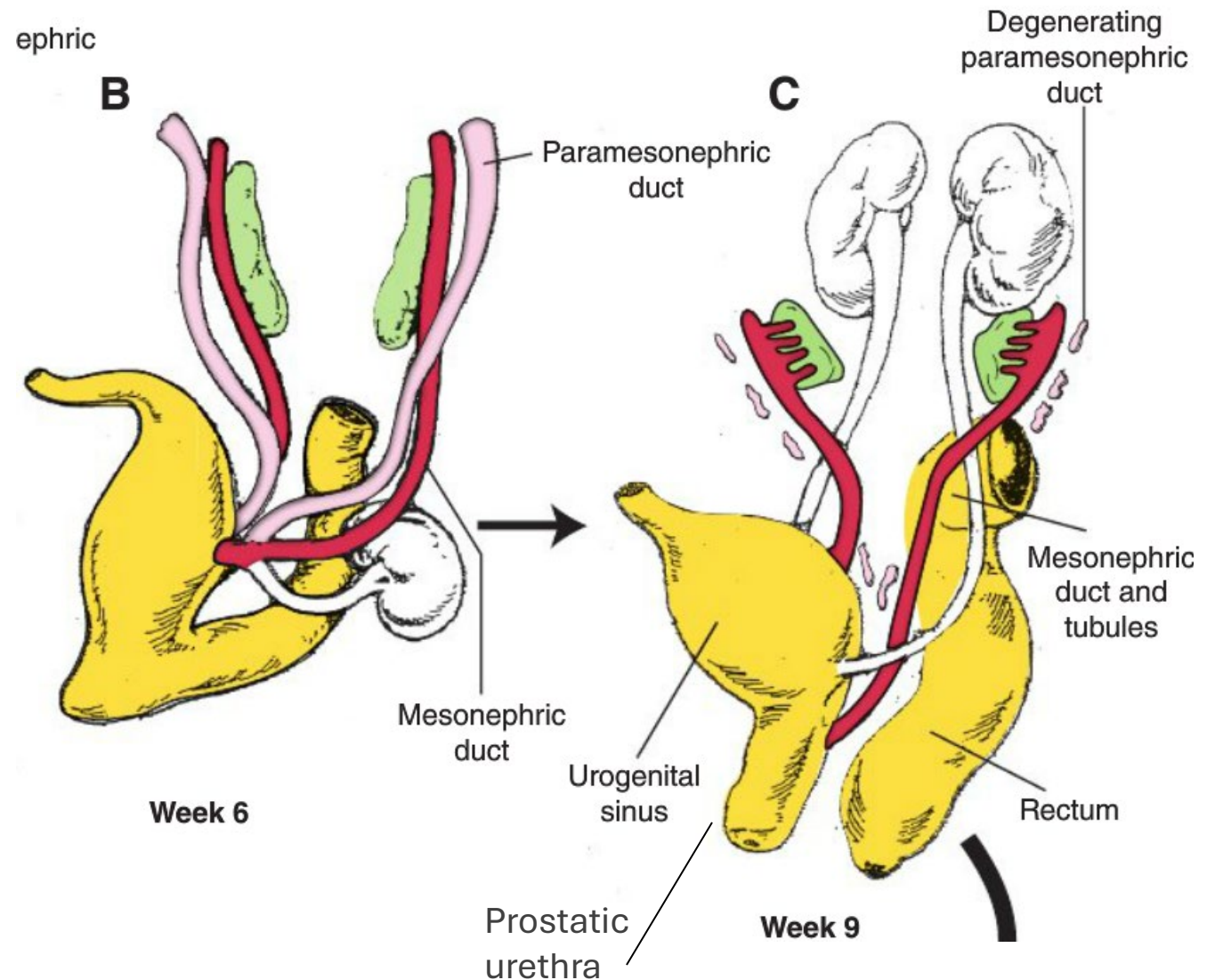
- A. Paramesonephric (Müllerian) duct
- B. Mesonephric (Wolffian) duct
- C. Urogenital sinus
- D. Genital tubercle
- E. Genital ridge

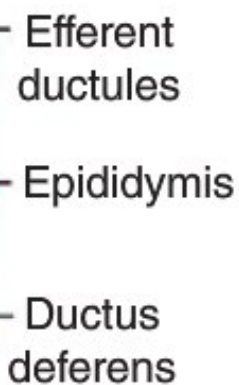
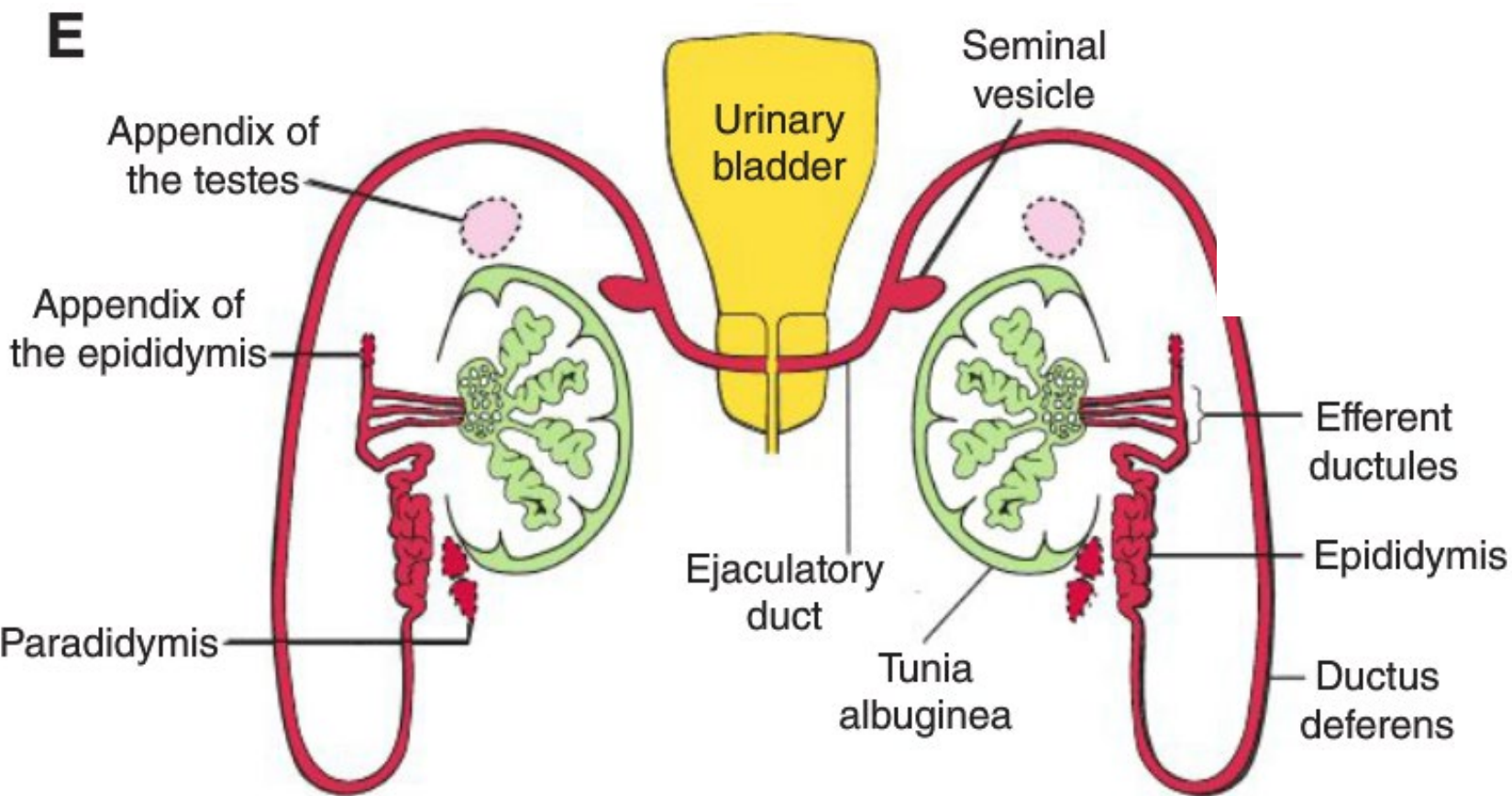


Development of the Male Reproductive Tract

- Wolffian duct (SEED= Seminal vesicle, Epididymis, Ejaculatory duct, Ductus deferens). In females, remnant: Gartner duct.
- Paramesonephric duct degenerates due to MIF by Sertoli cells of testes.
- The urogenital sinus:
 1. bladder.
 2. prostatic/membranous urethra.
 3. penile urethra.

The ureters enter the bladder (area of the Trigone), and the Wolffian ducts (vas deferens) enter into the prostatic urethra





Question 7

A 19-year-old male is evaluated for infertility. Physical examination is unremarkable. Semen analysis demonstrates markedly decreased sperm volume and absent fructose. Pelvic MRI reveals bilateral absence of the seminal vesicles, while the testes and epididymides are present and normally developed. The absent structures are derived from which embryologic structure?

- A. Paramesonephric (Müllerian) duct
- B. Mesonephric (Wolffian) duct
- C. Urogenital sinus
- D. Genital tubercle
- E. Genital ridge

B



Development of the Male Reproductive Tract

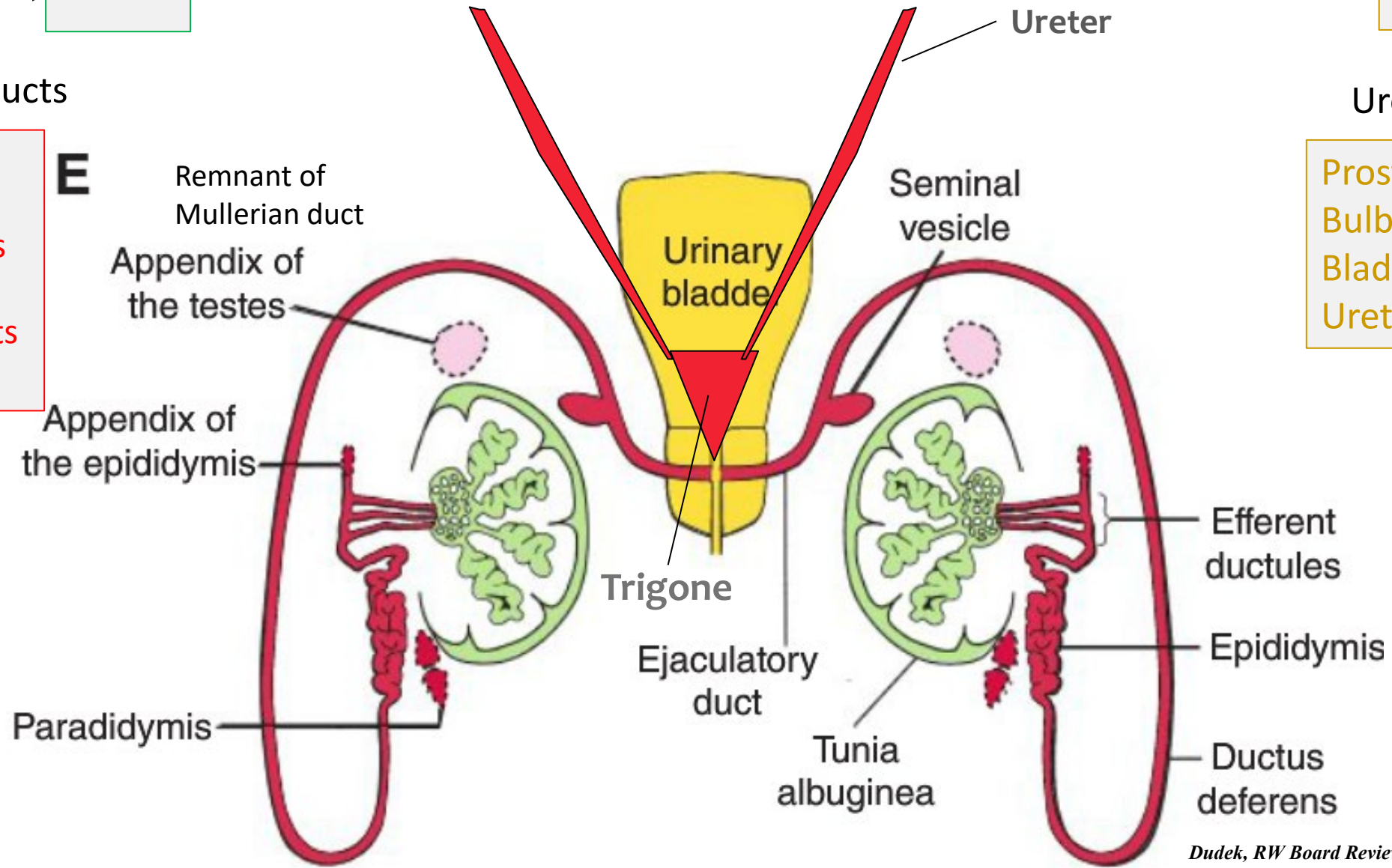
Mesoderm → testes

Mesonephric ducts

- Efferent ducts
- Epididymis
- Ductus deferens
- Seminal vesicle
- Ejaculatory ducts
- Trigone

Endoderm

- Urogenital sinus
- Prostate gland
- Bulbourethral gland
- Bladder
- Urethra



8



Question 8

A 6-year-old child with a 46, XY karyotype is evaluated for bilateral cryptorchidism. During laparoscopic orchiopexy, the surgeon unexpectedly identifies a uterus and fallopian tubes. Histologic examination confirms normal testes. Serum testosterone is normal for age. Failure of regression of which embryologic structure is responsible for the persistence of the additional reproductive organs found in this patient?

- A. Mesonephric ducts
- B. Genital tubercle
- C. Urogenital sinus
- D. Paramesonephric ducts
- E. Genital ridge

Sex Differentiation

SIGNAL	SOURCE	ACTION	RESULT IF ABSENT
SRY gene	Y chromosome	Induces indifferent gonad → testis (Sertoli + Leydig cells)	Gonad becomes ovary (default) NRE
AMH (Anti-Müllerian Hormone)	Sertoli cells of testis	Causes regression of Müllerian (paramesonephric) ducts	Müllerian ducts persist → uterus + tubes + upper vagina form NRE
Testosterone	Leydig cells of testis	Stabilises Wolffian (mesonephric) ducts → male reproductive tract	Wolffian ducts regress → no vas deferens, epididymis, seminal vesicles NRE
DHT (Dihydrotestosterone)	Testosterone converted by 5 α -reductase in target tissues	Masculinises external genitalia (penis, scrotum, prostate)	Female external genitalia form (default) — even in XY individuals NRE

Question 8

A 6-year-old child with a 46, XY karyotype is evaluated for bilateral cryptorchidism. During laparoscopic orchiopexy, the surgeon unexpectedly identifies a uterus and fallopian tubes. Histologic examination confirms normal testes. Serum testosterone is normal for age. Failure of regression of which embryologic structure is responsible for the persistence of the additional reproductive organs found in this patient?

- A. Mesonephric ducts
- B. Genital tubercle
- C. Urogenital sinus
- D. Paramesonephric ducts
- E. Genital ridge

D

Persistent Müllerian Duct Syndrome

Caused by deficiency of anti-Müllerian hormone (AMH) or defects in the AMH receptor.

Normally, Sertoli cells secrete AMH, causing regression of the paramesonephric (Müllerian) ducts.

The Müllerian ducts give rise to: fallopian tubes, uterus, cervix, upper vagina

Because testosterone production is normal, testes develop normally, and Wolffian derivatives are present. However, failure of Müllerian regression results in persistence of female internal reproductive structures.

Persistent Müllerian Duct Syndrome

46,XY male with uterus and fallopian tubes

Mechanism	AMH deficiency or AMH receptor mutation → Müllerian ducts persist in 46,XY male
Result	Phenotypically normal male with uterus and fallopian tubes inside the abdomen — testes present
Presents as	Male infant with inguinal hernia or cryptorchidism — uterus found in hernia sac at surgery
External	Normal male external genitalia — DHT and testosterone work normally
Treatment	Surgical removal of Müllerian remnants. Orchidopexy for undescended testes

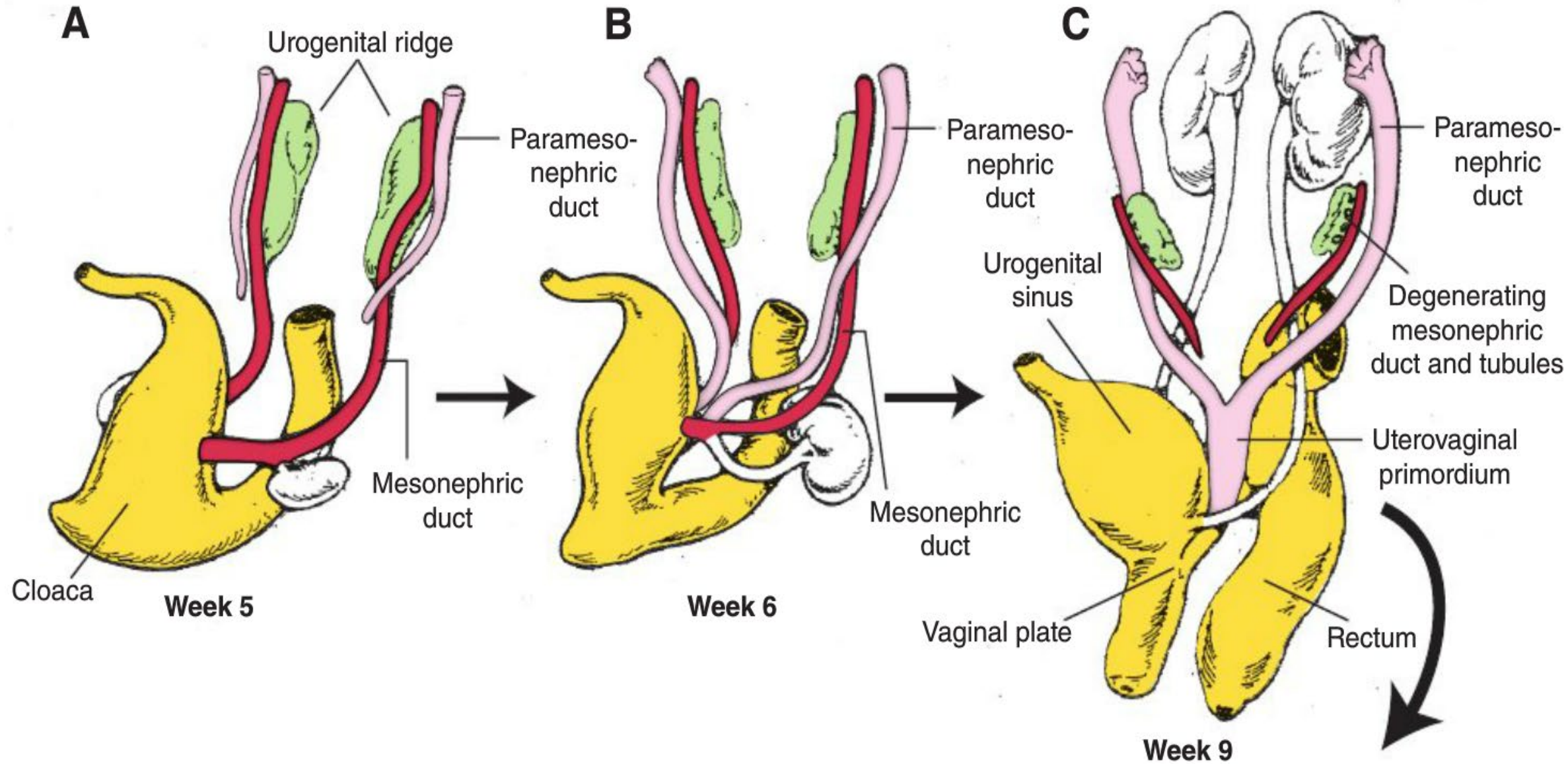
9

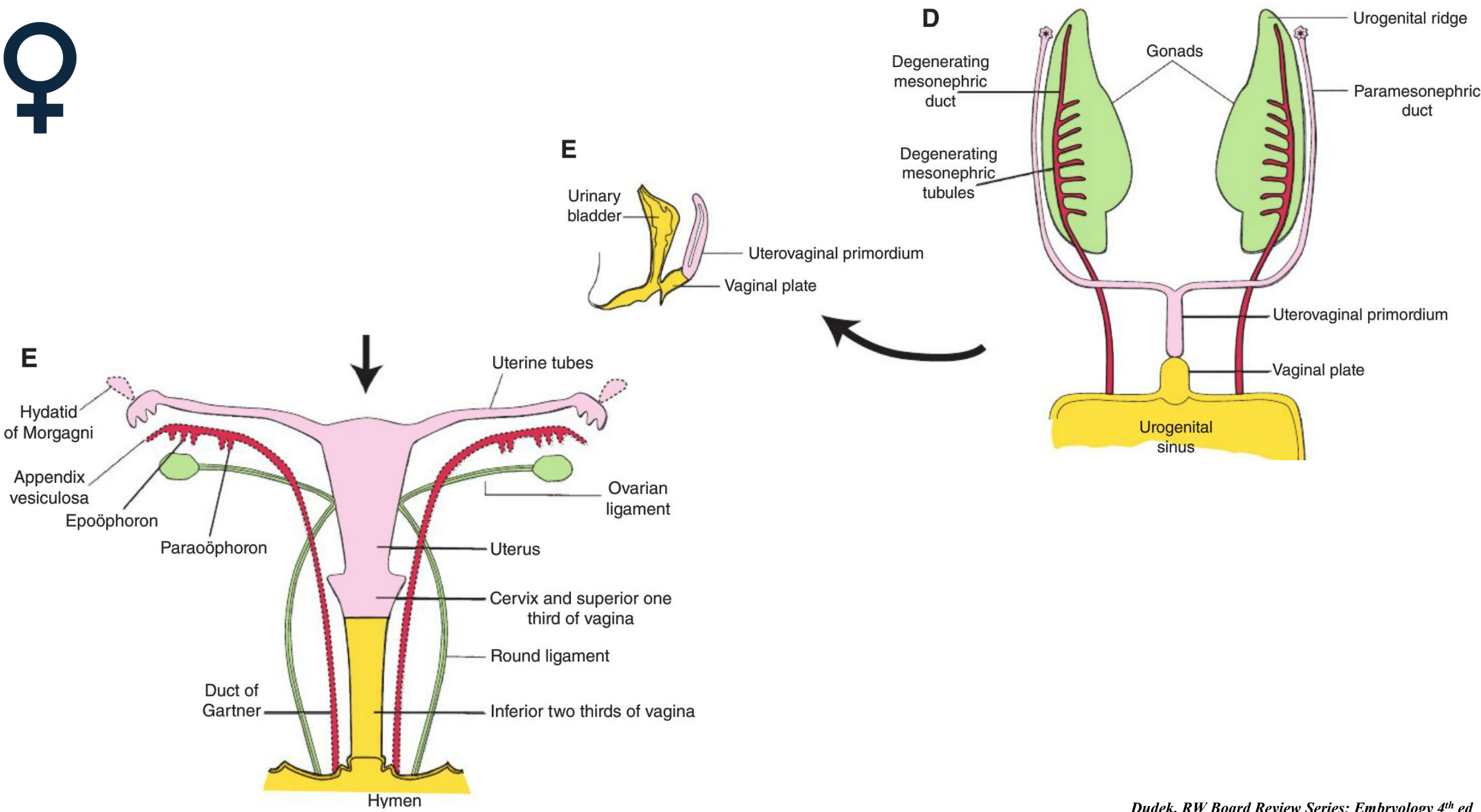


Question 9

A 26-year-old woman with a history of recurrent pregnancy loss undergoes hysterosalpingography. Imaging demonstrates two separate uterine horns, each connected to its own cervix. The ovaries appear normal bilaterally. The abnormality most likely resulted from failure of which developmental process?

- A. Differentiation of the gonadal ridge
- B. Fusion of the paramesonephric ducts
- C. Regression of the mesonephric ducts
- D. Formation of the urogenital sinus
- E. Migration of primordial germ cells





Question 9

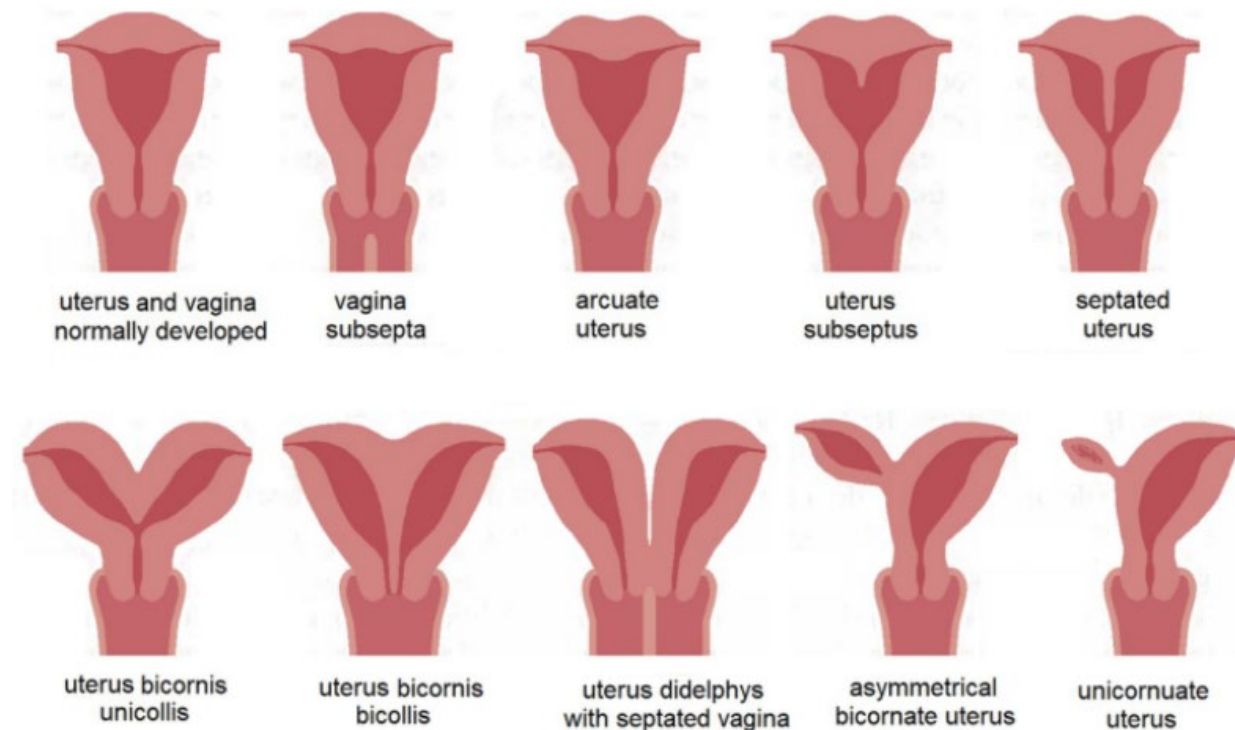
A 26-year-old woman with a history of recurrent pregnancy loss undergoes hysterosalpingography. Imaging demonstrates two separate uterine horns, each connected to its own cervix. The ovaries appear normal bilaterally. The abnormality most likely resulted from failure of which developmental process?

- A. Differentiation of the gonadal ridge
- B. Fusion of the paramesonephric ducts
- C. Regression of the mesonephric ducts
- D. Formation of the urogenital sinus
- E. Migration of primordial germ cells

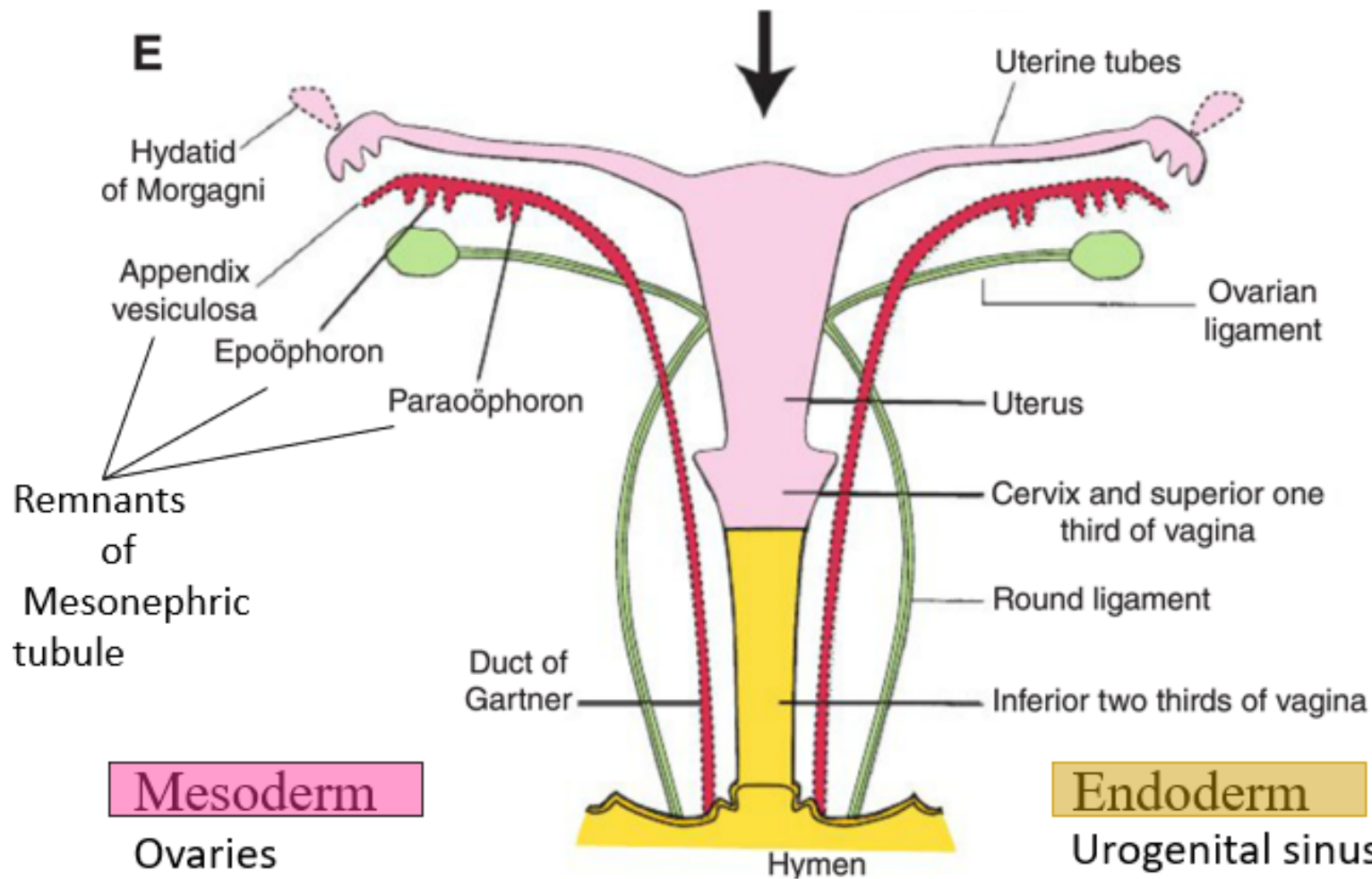
B

Uterine Abnormalities

Defect	Embryologic Abnormality
Uterus didelphys	Complete failure of Müllerian duct fusion
Bicornuate uterus	Partial failure of Müllerian duct fusion
Septate uterus	Failure of resorption of the uterine septum after fusion



Development of the Uterus and Uterine Tubes



Mesoderm

Ovaries

Müllerian ducts

Uterine tubes

Uterus, cervix

upper 1/3 vagina

Endoderm

Urogenital sinus

Bladder

lower 2/3 vagina

Urethra

Urethral glands

- The cranial portion of the Müllerian duct forms the uterine tubes and the caudal portions fuse together to form the uterus, cervix and upper 1/3 of the vagina.
- The rest of the vagina is from the urogenital sinus.
- The ovaries descend into the pelvis; the round ligament may help with descent.

10



Question 10

A 17-year-old adolescent is evaluated for primary amenorrhea. Physical examination reveals normal breast development, sparse pubic and axillary hair, and a blind-ending vaginal pouch. Pelvic U/S demonstrates the absence of a uterus and fallopian tubes. Laboratory studies show elevated serum testosterone levels within the normal male range. Karyotype analysis reveals 46,XY. Which of the following is the most likely diagnosis?

- A. Müllerian agenesis (Mayer-Rokitansky-Küster-Hauser syndrome)
- B. Androgen insensitivity syndrome (AIS)
- C. 5 α -reductase deficiency
- D. Cryptorchidism
- E. Congenital adrenal hyperplasia

Sex Differentiation

SIGNAL	SOURCE	ACTION	RESULT IF ABSENT
SRY gene	Y chromosome	Induces indifferent gonad → testis (Sertoli + Leydig cells)	Gonad becomes ovary (default) NRE
AMH (Anti-Müllerian Hormone)	Sertoli cells of testis	Causes regression of Müllerian (paramesonephric) ducts	Müllerian ducts persist → uterus + tubes + upper vagina form NRE
Testosterone	Leydig cells of testis	Stabilises Wolffian (mesonephric) ducts → male reproductive tract	Wolffian ducts regress → no vas deferens, epididymis, seminal vesicles NRE
DHT (Dihydrotestosterone)	Testosterone converted by 5 α -reductase in target tissues	Masculinises external genitalia (penis, scrotum, prostate)	Female external genitalia form (default) — even in XY individuals NRE

Takeaway

AIS = 46,XY + testes + AMH present + androgen receptor defect

- AMH → no uterus, no fallopian tubes, blind-ending vagina
- Androgen insensitivity → female external genitalia
- Testosterone aromatization → normal breast development
- Lack of androgen action → sparse pubic and axillary hair
- This combination of primary amenorrhea + normal breasts + sparse pubic hair + blind-ending vagina + absent uterus + 46,XY is the classic presentation of Complete Androgen Insensitivity Syndrome (CAIS)

Androgen Insensitivity Syndrome

- AIS is caused by a mutation in the androgen receptor, resulting in complete resistance to testosterone and dihydrotestosterone (DHT). Although the testes develop normally under the influence of the SRY gene, the body's tissues cannot respond to androgens.
- Sertoli cells produce anti-Müllerian hormone (AMH), causing regression of the Müllerian ducts and resulting in the absence of the uterus and fallopian tubes.
- Testosterone is aromatized to estrogen, leading to normal breast development.

Androgen Insensitivity Syndrome (AIS)

Complete — 46,XY phenotypically female

Mechanism	Mutation in androgen receptor gene → cells cannot respond to testosterone or DHT
Karyotype	46,XY — genetically male
Gonads	Testes present (in abdomen or inguinal canal) — SRY still works
Internal	No uterus, no fallopian tubes (AMH still works → Müllerian ducts regress). No male ducts (testosterone ignored)
External	Female phenotype — cells ignore DHT. Blind-ending vagina. No pubic/axillary hair (androgens needed for hair)
Presents as	Primary amenorrhoea in phenotypically female adolescent. Inguinal hernia in infant (testes)

Question 10

A 17-year-old adolescent is evaluated for primary amenorrhea. Physical examination reveals normal breast development, sparse pubic and axillary hair, and a blind-ending vaginal pouch. Pelvic U/S demonstrates the absence of a uterus and fallopian tubes. Laboratory studies show elevated serum testosterone levels within the normal male range. Karyotype analysis reveals 46,XY. Which of the following is the most likely diagnosis?

- A. Müllerian agenesis (Mayer-Rokitansky-Küster-Hauser syndrome)
- B. Androgen insensitivity syndrome (AIS)
- C. 5 α -reductase deficiency
- D. Cryptorchidism
- E. Congenital adrenal hyperplasia

B

11



Question 11

A child is assigned female at birth because of female-appearing external genitalia. Over the next several years, the child prefers traditionally male activities and experiences difficulty fitting in with female peers. At age 12, puberty is marked by rapid enlargement of the phallus, increased muscle mass, deepening of the voice, and male-pattern body hair growth. Physical examination reveals a blind-ending vaginal pouch and palpable gonads in the inguinal canals. Karyotype analysis demonstrates 46-XY. Which of the following is the most likely cause of this patient's condition?

- A. Androgen insensitivity syndrome
- B. Congenital adrenal hyperplasia
- C. 5 α -reductase deficiency
- D. Persistent Müllerian duct syndrome
- E. Gonadal dysgenesis

Sex Differentiation

SIGNAL	SOURCE	ACTION	RESULT IF ABSENT
SRY gene	Y chromosome	Induces indifferent gonad → testis (Sertoli + Leydig cells)	Gonad becomes ovary (default) NRE
AMH (Anti-Müllerian Hormone)	Sertoli cells of testis	Causes regression of Müllerian (paramesonephric) ducts	Müllerian ducts persist → uterus + tubes + upper vagina form NRE
Testosterone	Leydig cells of testis	Stabilises Wolffian (mesonephric) ducts → male reproductive tract	Wolffian ducts regress → no vas deferens, epididymis, seminal vesicles NRE
DHT (Dihydrotestosterone)	Testosterone converted by 5α-reductase in target tissues	Masculinises external genitalia (penis, scrotum, prostate)	Female external genitalia form (default) — even in XY individuals NRE

Key Diagnostic Clue

46,XY + female/ambiguous genitalia at birth + virilization at puberty

Explanation

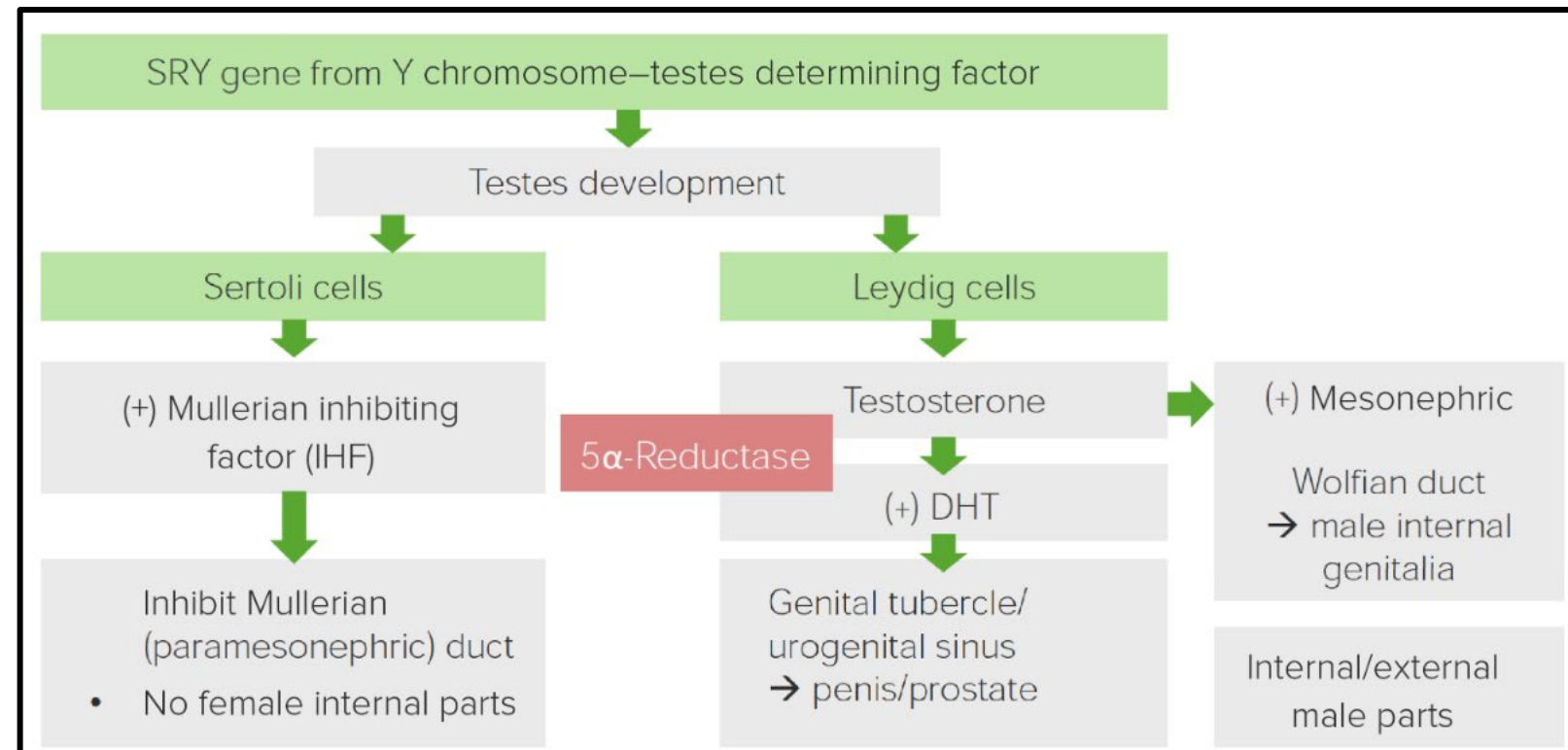
- During fetal development, testosterone must be converted to dihydrotestosterone (DHT) by the enzyme 5 α -reductase. DHT is required for the development of the penis, scrotum, and prostate
- When DHT is deficient, external genitalia appear female or ambiguous at birth. However, at puberty, rising testosterone levels can produce significant masculinization.

"Virilizes Later" = 5 α -Reductase Deficiency

- DHT is needed for fetal external genitalia
- Testosterone drives pubertal virilization

"Androgens Can't Act" = AIS

- Testosterone present
- Receptors absent
- No virilization despite normal/high testosterone levels



Question 11

A child is assigned female at birth because of female-appearing external genitalia. Over the next several years, the child prefers traditionally male activities and experiences difficulty fitting in with female peers. At age 12, puberty is marked by rapid enlargement of the phallus, increased muscle mass, deepening of the voice, and male-pattern body hair growth. Physical examination reveals a blind-ending vaginal pouch and palpable gonads in the inguinal canals. Karyotype analysis demonstrates 46-XY. Which of the following is the most likely cause of this patient's condition?

- A. Androgen insensitivity syndrome
- B. Congenital adrenal hyperplasia
- C. 5 α -reductase deficiency
- D. Persistent Müllerian duct syndrome
- E. Gonadal dysgenesis

C

12



Question 12

A 2-week-old male infant is brought for a routine newborn examination. The left testis is palpated within the scrotum, but the right testis is not palpable in either the scrotum or inguinal canal. Ultrasonography confirms an undescended intra-abdominal right testis. The parents are counseled regarding the potential long-term consequences if the condition is not corrected. Which of the following is the most important long-term complication associated with this condition?

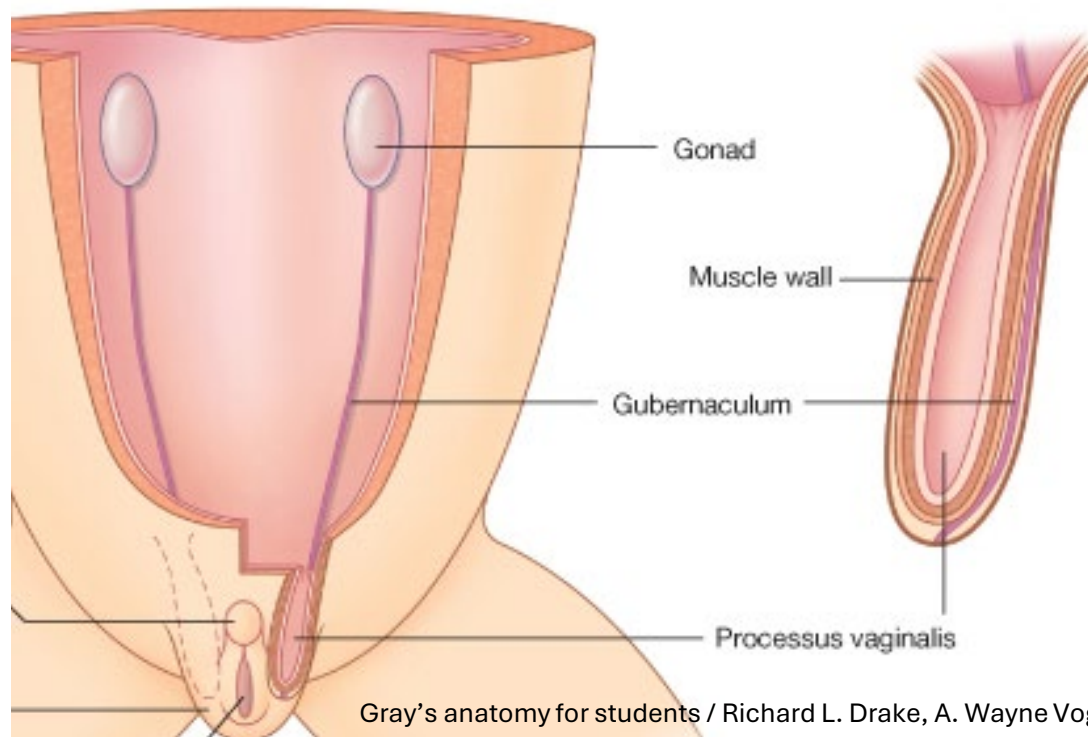
- A. Infertility due to heat-induced damage to both testes
- B. Hypospadias resulting from abnormal urethral development
- C. Increased risk of testicular germ cell malignancy (especially seminoma)
- D. Primary adrenal insufficiency due to impaired androgen production
- E. Persistent Müllerian duct syndrome

Descent of testis

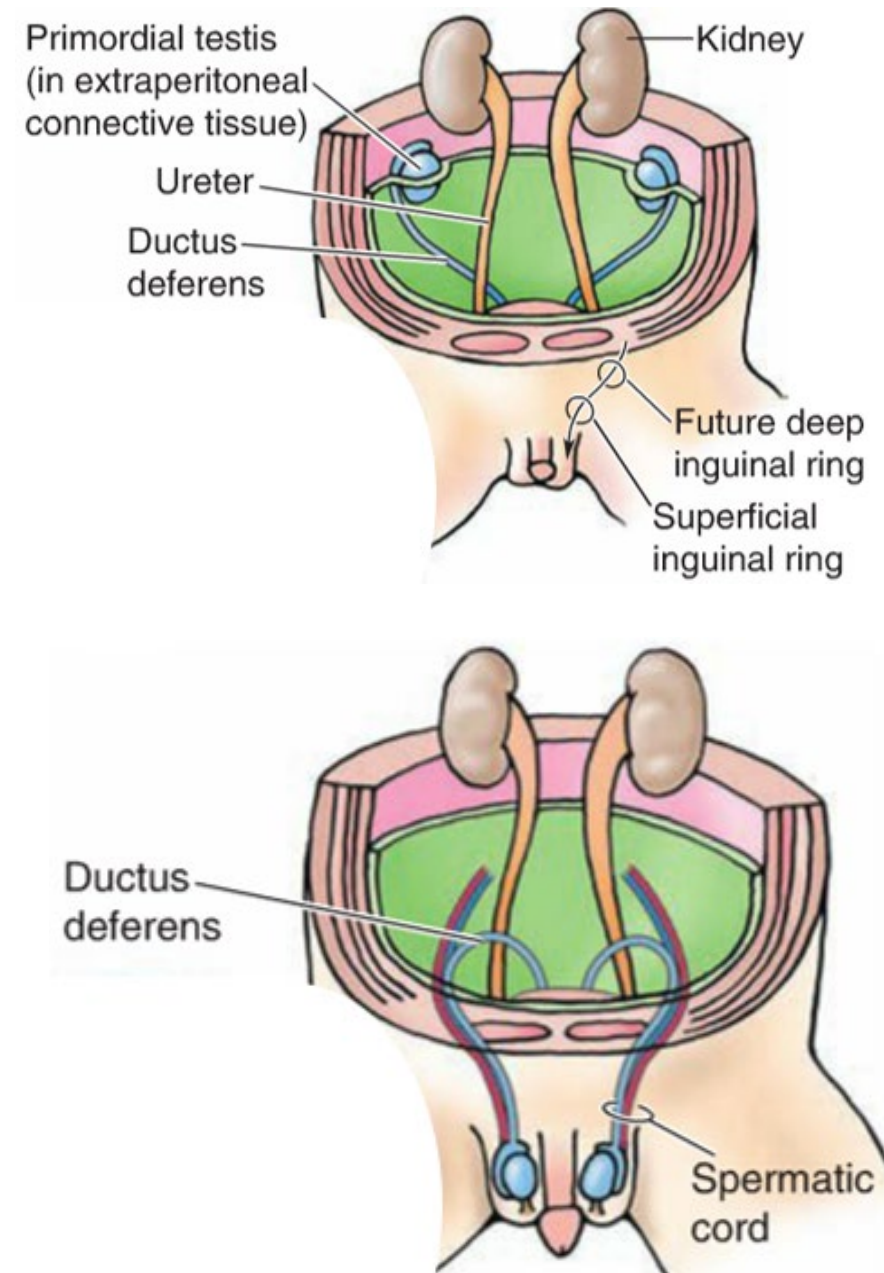
Inguinal canal is formed in relation to the relocation of the gonads during the fetal development.

Gubernaculum: is a fibrous band extend from the caudal part of testis. It plays a role in the descent of the testis.

Processus vaginalis: is a diverticulum or out pouching of the peritoneum.

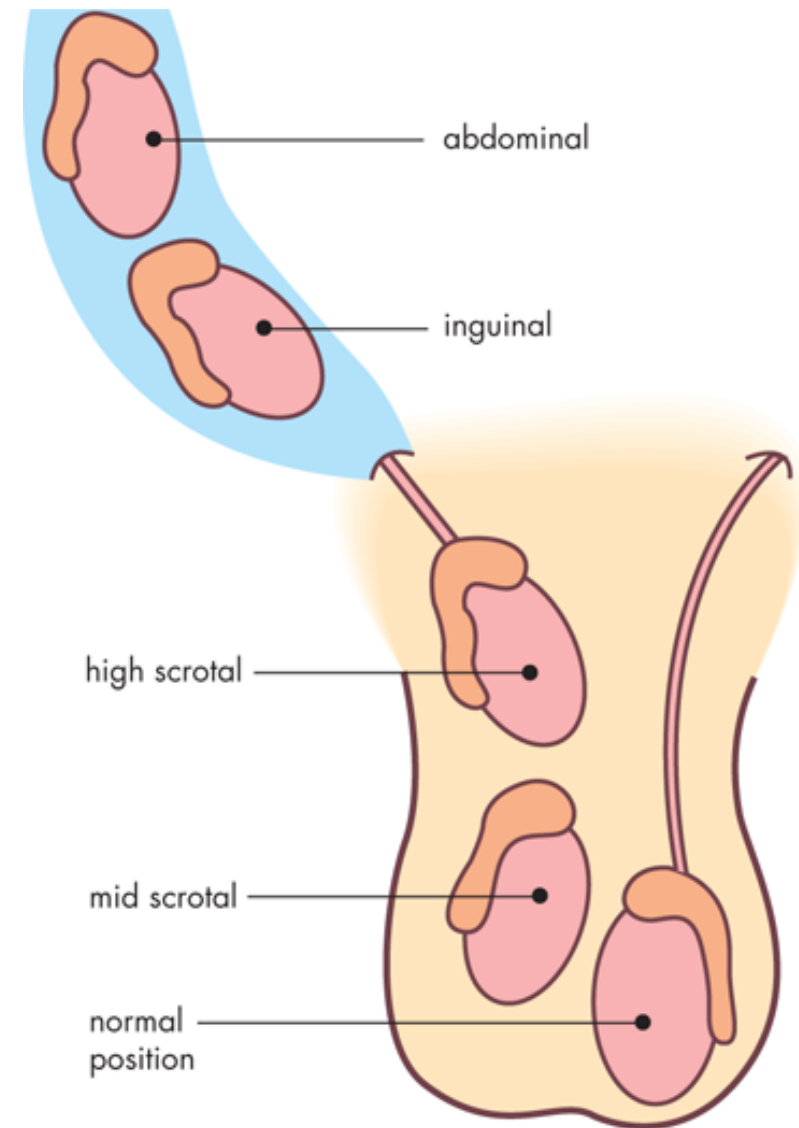


Gray's anatomy for students / Richard L. Drake, A. Wayne Vogl, & Adam W. M. Mitchell—2nd Ed.



Explanation

- This infant has cryptorchidism (undescended testis). Normally, the testes descend into the scrotum during late fetal development. Failure of descent results in an undescended testis, most commonly located in the inguinal canal or abdomen.
- In 1 % of newborns, one or both testes fail to descend into the scrotum. Undescended testes fail to produce mature sperm. The most important long-term complication of untreated cryptorchidism is an increased risk of testicular germ cell tumors, particularly seminoma.
- Cryptorchidism is also associated with:
 - ✓ Infertility or subfertility
 - ✓ Testicular atrophy
 - ✓ Increased risk of testicular torsion
 - ✓ Inguinal hernia



Source: John Murtagh, Jill Rosenblatt: *John Murtagh's General Practice*, 6e: www.murtagh.mhmedical.com
Copyright © McGraw-Hill Education. All rights reserved.

Question 12

A 2-week-old male infant is brought for a routine newborn examination. The left testis is palpated within the scrotum, but the right testis is not palpable in either the scrotum or inguinal canal. Ultrasonography confirms an undescended intra-abdominal right testis. The parents are counseled regarding the potential long-term consequences if the condition is not corrected. Which of the following is the most important long-term complication associated with this condition?

C

- A. Infertility due to heat-induced damage to both testes
- B. Hypospadias resulting from abnormal urethral development
- C. Increased risk of testicular germ cell malignancy (especially seminoma)
- D. Primary adrenal insufficiency due to impaired androgen production
- E. Persistent Müllerian duct syndrome

13



Question 13

During an anatomy lab session, a medical student learns that in females, a certain embryonic structure persists as the ovarian ligament and the round ligament of the uterus. These ligaments play an essential role in maintaining the anatomical position of the ovaries and uterus within the pelvis. Which structure gives rise to both of these ligaments in female development?

- A. Paramesonephric ducts
- B. Urogenital sinus
- C. Genital tubercle
- D. Gubernaculum
- E. Labioscrotal swellings

Question 13

During an anatomy lab session, a medical student learns that in females, a certain embryonic structure persists as the ovarian ligament and the round ligament of the uterus. These ligaments play an essential role in maintaining the anatomical position of the ovaries and uterus within the pelvis. Which structure gives rise to both of these ligaments in female development?

- A. Paramesonephric ducts
- B. Urogenital sinus
- C. Genital tubercle
- D. Gubernaculum
- E. Labioscrotal swellings

D

14

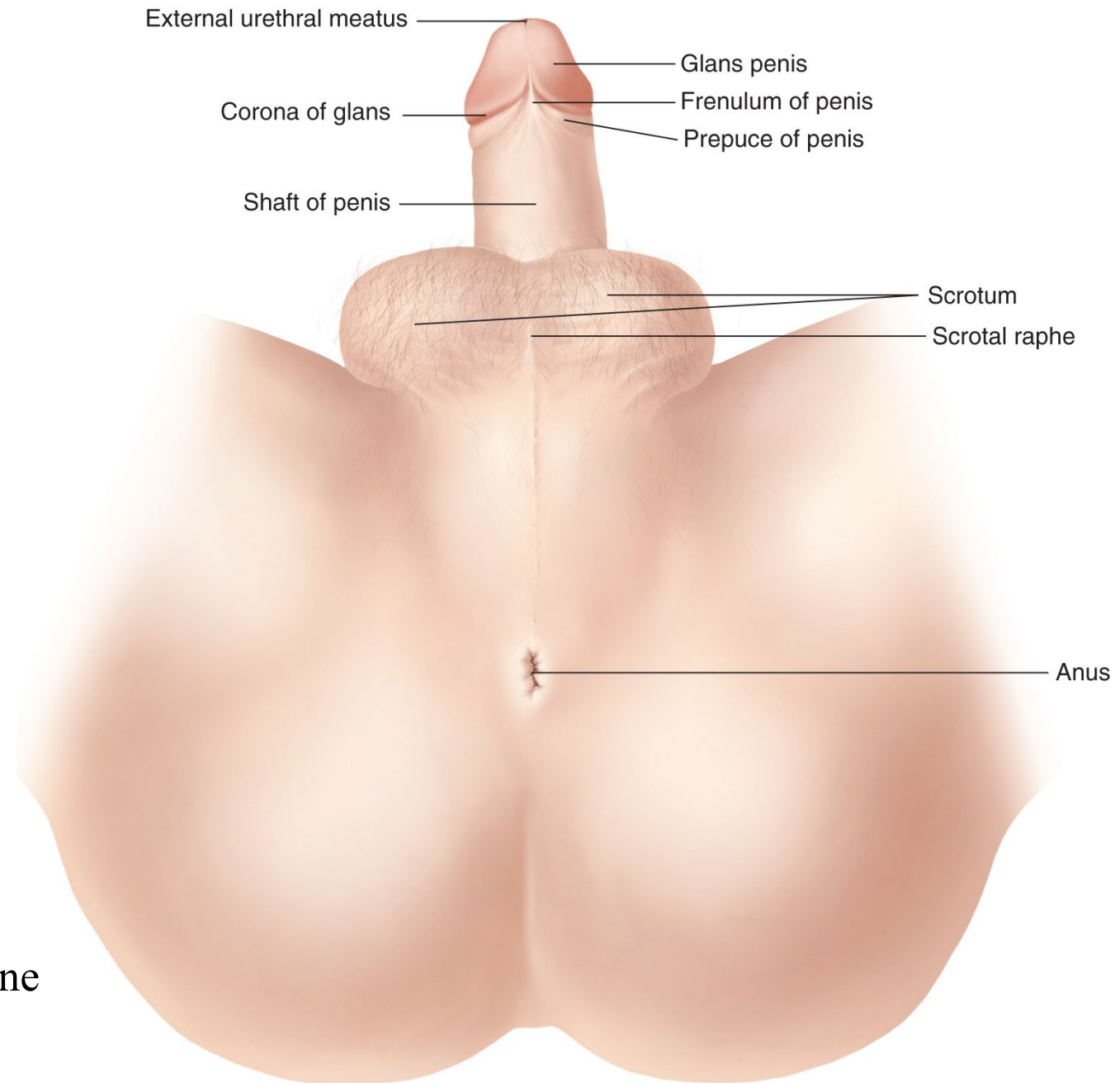
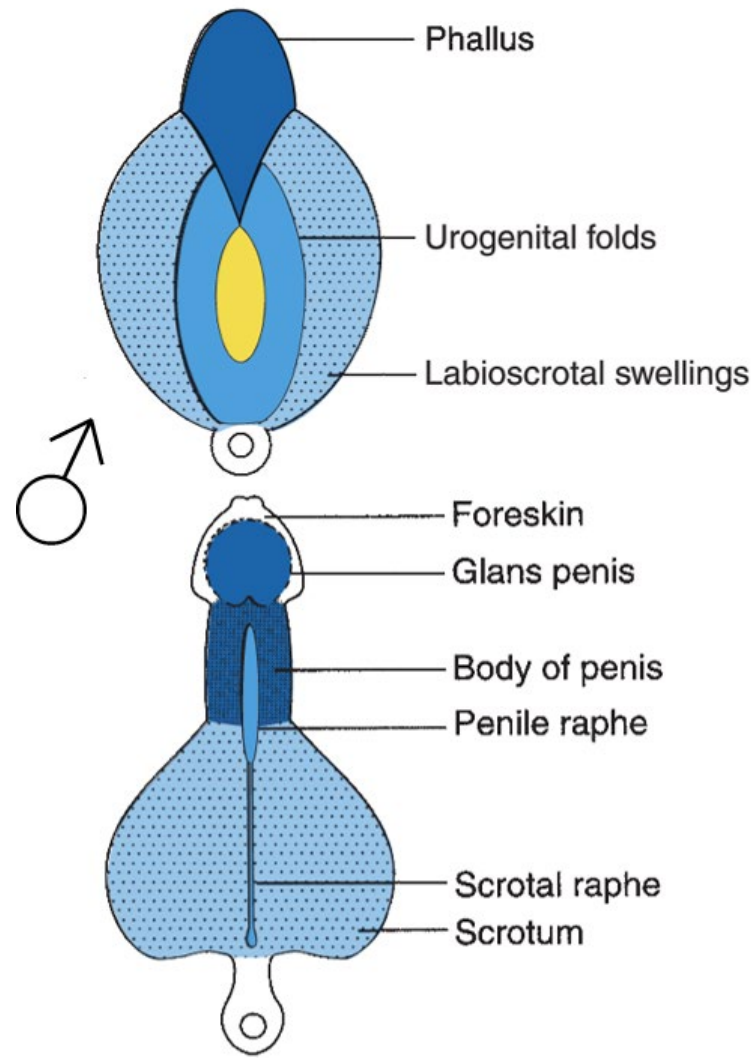


Question 14

A full-term male infant is noted at birth to have an abnormally positioned urethral opening on the ventral surface of the penis near the penoscrotal junction. Physical examination also demonstrates ventral curvature of the penis (chordee). The parents ask whether routine neonatal circumcision can be performed before discharge. Which of the following is the most appropriate recommendation?

- A. Circumcision can be performed routinely at 6 to 8 weeks of age.
- B. Circumcision should be delayed until puberty.
- C. Circumcision should not be performed because the foreskin may be required for definitive surgical repair.
- D. Circumcision will help correct the abnormal location of the urethral opening.
- E. Circumcision should be performed immediately to reduce the risk of urinary obstruction

Male Surface Anatomy



Proliferation of mesoderm around the cloacal membrane causes the overlying ectoderm to form three elevated structures. Stimulated by testosterone

Question 14

A full-term male infant is noted at birth to have an abnormally positioned urethral opening on the ventral surface of the penis near the penoscrotal junction. Physical examination also demonstrates ventral curvature of the penis (chordee). The parents ask whether routine neonatal circumcision can be performed before discharge. Which of the following is the most appropriate recommendation?

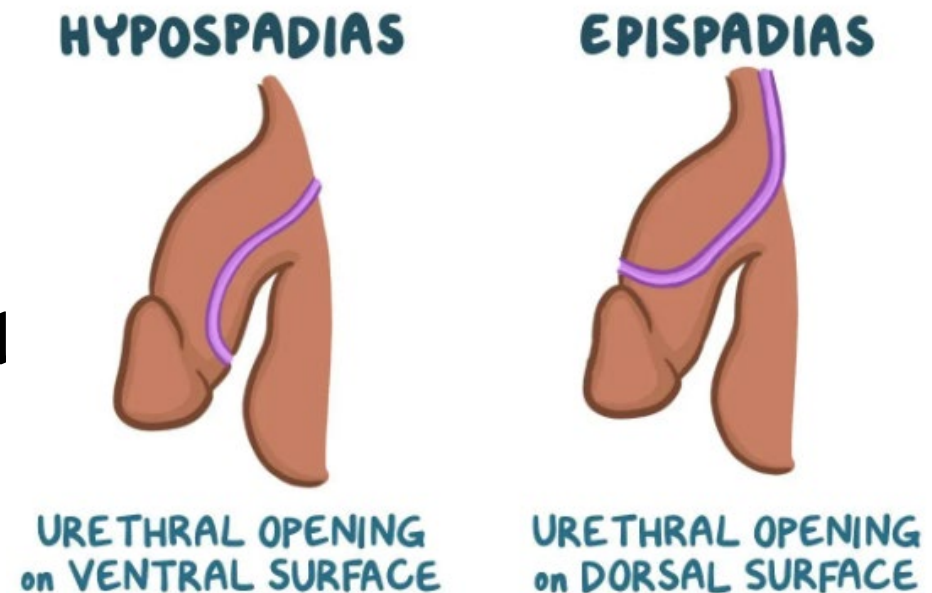
- A. Circumcision can be performed routinely at 6 to 8 weeks of age.
- B. Circumcision should be delayed until puberty.
- C. Circumcision should not be performed because the foreskin may be required for definitive surgical repair.
- D. Circumcision will help correct the abnormal location of the urethral opening.
- E. Circumcision should be performed immediately to reduce the risk of urinary obstruction

C

Explanation

- This infant has hypospadias, a congenital anomaly in which the urethral opening is located on the ventral surface of the penis due to incomplete fusion of the urethral folds. A key management principle is that circumcision should be avoided because the foreskin is frequently used as graft material during surgical reconstruction of the urethra. Therefore, preservation of the foreskin is essential until evaluation by a pediatric urologist.

- **Hypospadias:** Ventral urethral opening, due to incomplete fusion of urethral folds
- **Epispadias:** Dorsal urethral opening & associated with bladder exstrophy



15



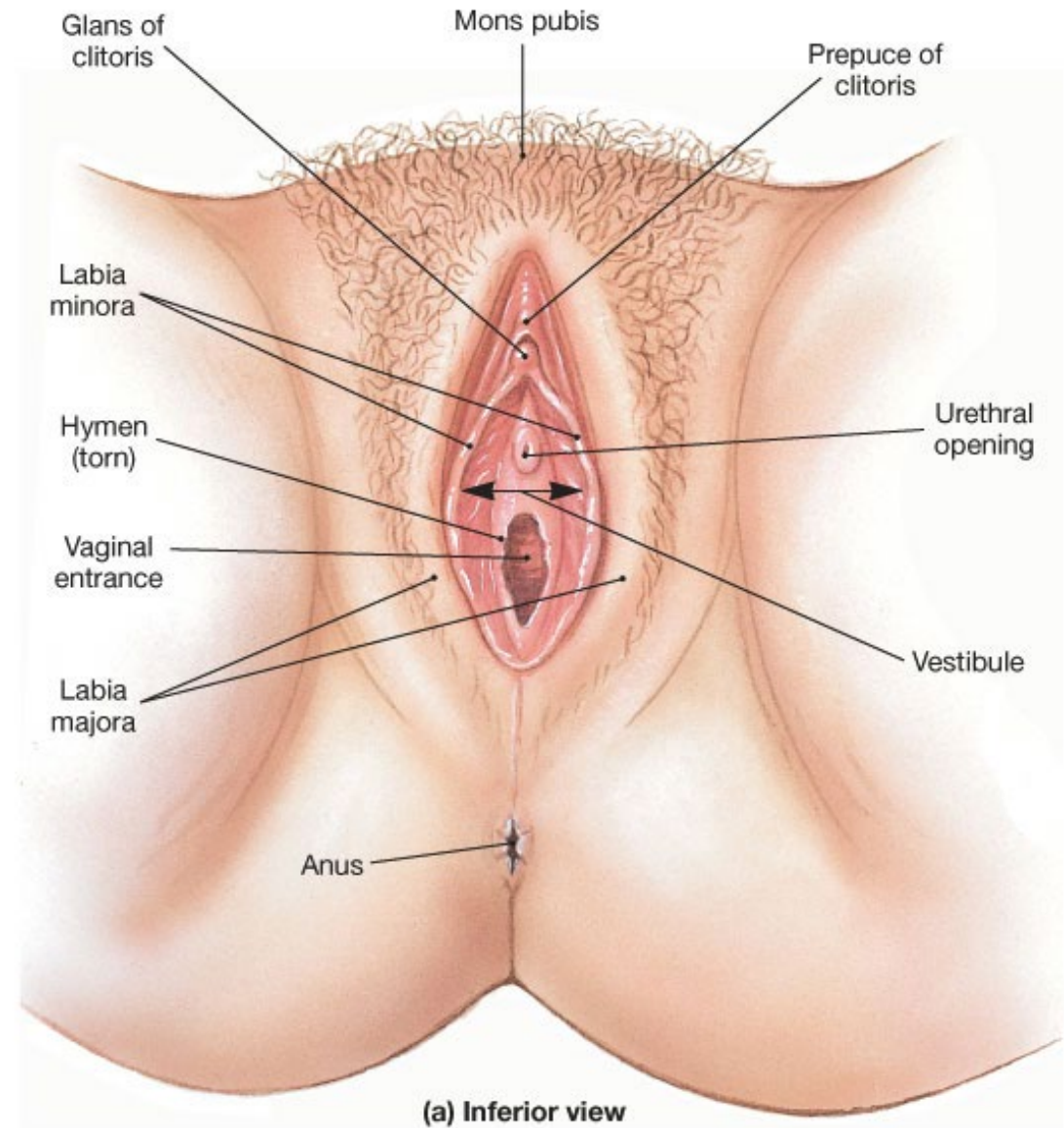
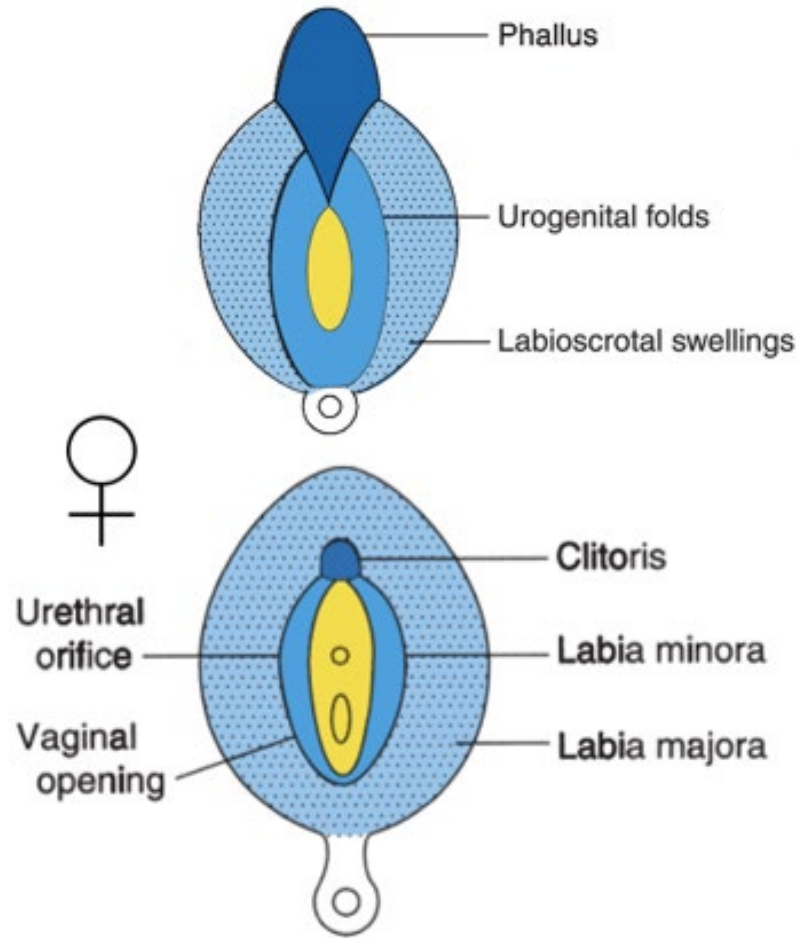
Question 15

A researcher is studying the embryologic development of the external genitalia. During the indifferent stage of development, paired labioscrotal swellings are present in both male and female embryos. In the absence of significant androgen stimulation, these structures develop into a normal female anatomical structure. Which of the following structures is derived from the labioscrotal swellings in a genetically normal 46, XX fetus?

- A. Clitoris
- B. Labia minora
- C. Vestibule of vagina
- D. Greater vestibular (Bartholin) glands
- E. Labia majora



Female Surface Anatomy



Question 15

A researcher is studying the embryologic development of the external genitalia. During the indifferent stage of development, paired labioscrotal swellings are present in both male and female embryos. In the absence of significant androgen stimulation, these structures develop into a normal female anatomical structure. Which of the following structures is derived from the labioscrotal swellings in a genetically normal 46, XX fetus?

- A. Clitoris
- B. Labia minora
- C. Vestibule of vagina
- D. Greater vestibular (Bartholin) glands
- E. Labia majora

E

Summary of ♀ & ♂ Reproductive system

** Remnants

Summary Table of Female and Male Reproductive System Development

Adult Female	Indifferent Embryo	Adult Male
Ovary, ovarian follicles, rete ovarii	Gonads mesoderm	Testes, seminiferous tubules, tubuli recti, rete testes, Leydig cells, Sertoli cells
Uterine tubes, uterus, cervix, superior one third of vagina <i>Hydatid of Morgagni</i> **	Paramesonephric duct mesoderm	<i>Appendix testes</i> **
<i>Appendix vesiculosa, Gartner duct</i> **	Mesonephric duct mesoderm	Epididymis, ductus deferens, seminal vesicle, ejaculatory duct
<i>Epoophoron, paroophoron</i> **	Mesonephric tubules mesoderm	Efferent ductules <i>Paradidymis</i> **
Glans clitoris, corpora cavernosa clitoris, vestibular bulbs	Genital tubercle	Glans penis, corpora cavernosa penis, corpus spongiosum
Labia minora	Urogenital folds	Ventral aspect of penis
Labia majora, mons pubis	Labioscrotal swellings	Scrotum
Ovarian ligament, round ligament of uterus	Gubernaculum	Gubernaculum testes
	Processus vaginalis	Tunica vaginalis
Inferior two thirds of vagina, glands	Urogenital sinus endoderm	Urethra, prostate gland, bulbourethral gland

The background of the image is a close-up of an ancient Egyptian stone wall. It features several rows of hieroglyphs carved into the light-colored, textured stone. The hieroglyphs are arranged in vertical columns. Some of the visible symbols include a lotus flower, a papyrus plant, a falcon, and various geometric shapes. The lighting is warm, highlighting the relief of the carvings.

THANK YOU